

CHAPTER

4

EMERGENCY PLANNING AND PREPAREDNESS

General Comments

This chapter is an expansion of the provisions found in the legacy fire codes used to develop the code. The overall approach has been to place all similar requirements into general sections.

This chapter first provides general scope and requirements for reporting emergencies and preventing interference with fire department activities in Section 401.

Section 403 provides detailed requirements based on occupancy. These requirements include seating plans for Group A occupancies. This section also establishes requirements for general public safety regarding fire watch personnel, crowd managers and planning requirements for public assemblages.

Section 404 provides detailed requirements for fire safety plans, fire evacuation plans and lockdowns.

The frequency and required documentation related to evacuation drills are addressed in Section 405. The minimum criteria for the training of occupants for emergency situations are found in Section 406. Section 407 provides requirements that apply to occupancies that contain hazardous materials. Among of the key elements are Hazardous Materials Inventory Statements (HMIS) and Hazardous Materials Management Plans (HMMP).

Purpose

More so than other fire-related requirements, this chapter focuses on the actions of occupants during fires and other emergencies. These requirements are based on higher levels of care necessary related to the concentration of people, physical and mental capabilities of the occupants, lack of familiarity with a building, or the complexity and size of the building. These requirements are intended to improve the effectiveness of other measures required by this code and the *International Building Code*® (IBC®).

Basically, this chapter addresses human contributions to life safety in buildings when a fire or other emergency occurs. The requirements for continuous training and scheduled evacuation drills can be as important as the required periodic inspections and maintenance of built-in fire protection features. The level of preparation by the occupants also improves the emergency responders' abilities during an emergency.

The IBC focuses on built-in fire protection features, such as automatic sprinkler systems, fire-resistance-rated construction and properly designed egress systems. The human element is only indirectly addressed in the IBC, whereas this chapter addresses the human element. This is a topic more traditionally addressed in fire codes along with the maintenance and use of buildings and systems.

Properly equipping and maintaining buildings to prevent damage and loss of life in the event of a fire involves two complementary fire safety strategies: managing the fire and managing the occupants. These strategies are discussed in more detail in the commentary to Chapter 1.

Managing Fire

A fire can be either prevented or managed. This chapter of the code focuses on training and preparedness while also emphasizing prevention. In some cases, moving occupants to minimize their exposure to a hazard is difficult or impractical. In these situations, controlling or eliminating the hazard is preferable, especially while it is still manageable. In fact, this is the concept underlying all fire-suppression requirements. Successful fire control depends on building occupants recognizing the fire threat, deciding to respond, choosing how to respond and, in the case of choosing fire control, identifying, locating and using the correct method. All of these functions must promptly take place in that order. Failure to perform promptly may preclude alternative strategies; therefore, location and identification of fire extinguishers and occupant standpipe hose lines are provided so that incipient firefighting equipment is readily accessible to occupants. These appliances, however, are often difficult to operate, and regardless of experience, fighting fire is a difficult and dangerous task. This chapter prescribes training requirements that, if met, will better prepare occupants to respond to incipient fires.

Managing Occupants

The management of occupants is primarily moving them away from the hazard. Verifying that enough exits have ample capacity and are immediately available, adequately arranged, appropriately identified and suitably protected are only the first steps toward achieving functional life safety. Occupants must know not only where exits are, but also when and how to use them. For instance, studies have shown that people have a "learned irrelevance" to emergency exits. Learned irrelevance is a psychological phenomenon that occurs when a person is exposed to a stimulus but usually does not need to

respond to it. Because of this phenomenon, most occupants are likely to exit the way they have entered, whether it is the correct way or not; therefore, beyond designing the building with an adequate number of exits, a method of encouraging the use of the best exits must be developed. Identifying dangerous conditions, deciding how to act and responding appropriately and promptly are essential. Various factors and situations can make evacuation not only difficult but potentially impractical. All of these factors involve the interaction between the building, its systems and features, occupants and the fire. This code concentrates on the last two factors while the IBC regulates the first two factors. Planning for life safety requires a response to these factors by defining the life safety strategies that must be implemented, as well as the means to achieve them. Life safety factors, such as buildings, fire and people, are important in managing exposed occupants.

This chapter concentrates on planning and practicing the desired actions of building occupants when a fire occurs. The remainder of the code focuses on the behaviors and procedures that must be practiced or observed to prevent or control a fire. The best way to create a safe building environment is through fire prevention. No system can ensure complete protection of building occupants.

Fires are not the only emergencies necessitating the implementation of life safety strategies; therefore, this chapter describes requirements for preparing and implementing life safety plans and programs in occupancies with special life safety problems. These include occupancies in which the number of occupants or the arrangement or complexity of the building may make evacuation or removal from hazardous conditions difficult or impractical.

A reality of the twenty-first century in the wake of terrorist attacks and school and business shootings is a need to establish so-called “lockdown” procedures in order to protect and defend building occupants. While the code does not mandate the implementation of lockdown plans and procedures, it does provide some minimum standards such plans must meet when they are prepared.

Not all occupants of each building are equally capable of performing tasks essential to their safety. A growing awareness, not only of people with physical disabilities, but also of what constitutes a disability, has focused life safety on everyone. Federal health care policies and funding criteria have spurred the deinstitutionalization of people who were formerly confined to nursing homes and other traditional health care institutions. This has created a new category of occupancies—board and care homes (institutional, residential care)—while the number of beds provided in nursing homes and hospitals continues to grow. Similarly, technological advances have promoted the creation of larger and more complex buildings, including high rises, open malls, domed stadiums, mixed-use complexes and convention centers. All of these situations create special life safety problems that physical features alone cannot remedy. Additionally, these situations require not only that adequate physical accommodations be provided, but also that building occupants be trained to respond to emergencies in these facilities.

Life Safety

Life safety strategies involve the development of an explicit statement of a desired life safety outcome. This statement, once designed to the capabilities of the building occupants and the physical arrangement of the exits, becomes a life safety strategy. Such approaches stress defining a specific strategy or strategies for protecting occupants. Protection may include moving them (assisting), causing them to move (directing), defending them in place or a combination of these measures. An effective strategy must consider the number and capabilities of building occupants; the type, location and arrangement of building exits; the fire; its effects on the people and building; and the number, training and capability of staff to direct or perform fire evacuation or incipient firefighting duties. Each strategy, combined with effective planning and practice, becomes the means for achieving the desired life safety outcome.

The life safety strategies for a health care facility, a high-rise office building and a multiplex theater could vary considerably based on the specific characteristics of the use. First, while the number of occupants will be significant in each case, the actual number occupying the building may be varied. Similarly, the occupant density and location of people in the building will vary, as well as the physical arrangement of the building, which in the first example that follows may be assumed as primarily horizontal and, in the second, as principally vertical. The largest difference will be the capabilities of the occupants. In high rises and theaters, building occupants will be expected to perform life safety behaviors themselves, while care recipients in a health care facility may require substantial assistance or need to defend in place. High-rise and theater occupants will differ from each other in their levels of familiarity with the building design. Further, in a theater, lighting conditions may interfere with the occupants’ ability to discern the path of egress travel.

In the first two examples, health care and high rise, removing all occupants from the building in the event of fire is impractical. In a high-rise building, occupants located above a fire are in greater danger than occupants located below the fire, since combustion products naturally rise. In a health care occupancy, the risk is compounded because many occupants may not be ambulatory or are generally incapable of self preservation. In both of these examples, a life safety strategy should first address the needs of those at greatest risk by removing them from harm. Secondly, the life safety strategy should stress separating occupants most at risk and their immediate neighbors from danger until the hazard can be controlled or confined. As seen in each example, life safety strategies should also incorporate both partial relocation and

defend-in-place concepts. In a health care facility, however, care recipients will typically be moved horizontally to achieve this objective, while in high-rise buildings, occupants will be expected to move downward or upward to separate themselves from danger. In the case of a multiplex theater, occupants will usually be directed to the nearest exit; however, its location may not be known to all or some of them. Further, because employees in assembly occupancies must be trained in the proper use of portable fire extinguishers, the life safety strategy should include instruction in using these appliances to minimize occupant exposure to fire effects. The resulting life safety strategies for these occupancies may resemble the following:

Example 1—Health Care Facility: Upon notice of fire, direct or assist evacuation and relocate occupants from the area of fire origin to an adjacent smoke compartment through horizontal exits.

Example 2—High-rise Building: Direct occupants in the area or floor of fire origin to the nearest exits. Occupants on the fire floor, the floor above and the floor below will relocate sequentially up or down at least two floors. Occupants located two floors above and two floors below the fire floor will be sequentially relocated following movement of fire floor occupants.

Example 3—Multiplex Theater: Announce exit locations and evacuation instructions prior to each movie. Over voice/alarm systems, direct occupants to nearest exit. Employees in the immediate vicinity of an incipient fire may attempt to control or extinguish it using a portable fire extinguisher after activating the fire alarm system.

Once the appropriate strategy has been defined, a plan can be expanded with little additional effort to form the backbone of a comprehensive life safety protocol. The following statements provide additional instructions for these examples:

Example 1—Health Care Facility: Monitor or reinforce fire barriers and smoke barriers so that they provide adequate defense against fire until it is controlled and extinguished. Staff will report progress of the fire and relocation operation to the Private Branch Exchange (PBX) operator through the nurse call station in adjacent smoke compartments.

Example 2—High-Rise Building: Building fire manager will meet fire department personnel at the fire command center located off the main lobby.

Example 3—Multiplex Theater: Projectionist will stop films so that the alarm and evacuation instructions are heard and followed. Upon activation of the alarm, instructions are heard and followed while the on-duty manager telephones the fire department to confirm that the fire was reported. Ushers will follow occupants out of each theater as conditions permit, closing exit doors and preventing reentry. Upon completing assigned duties, all staff will report to the manager located in front of the lobby entrance.

These expanded statements certainly do not constitute a fully developed plan; however, with these elements of the life safety plan defined, details can be added to a plan that addresses everything from preincident preparation through post-incident follow-up. A well-developed plan should include all or most of the following elements:

- Assignment of roles and responsibilities.
- Description of fire protection systems, including operating instructions, if appropriate.
- Building floor plans and sections.
- Seating diagrams and occupant load.
- Number, location and path of travel to exits.
- Emergency notification lists and procedures.
- Post-incident follow-up procedures, including salvage and insurance information.
- Plan revision and evaluation procedures.

Once a plan is developed, reviewed and approved, it must be distributed, practiced and periodically revised.

SECTION 401—GENERAL

401.1 Scope. Reporting of emergencies, coordination with emergency response forces, emergency plans and procedures for managing or responding to emergencies shall comply with the provisions of this section.

Exception: Firms that have *approved* on-premises firefighting organizations and that are in compliance with *approved* procedures for fire reporting.

C This section describes the overall scope of Chapter 4, which notes that all procedures relating to reporting and managing fire and other emergencies be in accordance with this chapter. There is one exception recognizing organizations, such as large industrial sites, that have on-site fire brigades. The fire brigades and the associated reporting procedures must be approved by the fire code official.

401.2 Approval. Where required by this code, fire safety plans, emergency procedures and employee training programs shall be *approved* by the fire code official.

C To verify that emergency procedures, training and fire safety plans have taken all essential factors into account, the plans and procedures must be approved by the fire code official.

EMERGENCY PLANNING AND PREPAREDNESS

401.3 Emergency responder notification. Notification of emergency responders shall be in accordance with Sections 401.3.1 through 401.3.3.

C This section simply states that the notification of emergency forces must comply with Section 401.3 and all of its subsections.

401.3.1 Fire events. In the event an unwanted fire occurs on a property, the *owner* or occupant shall immediately report such condition to the fire department.

C This section requires prompt notification of the fire department in the event of a fire emergency. Employees or other occupants are prohibited from delaying in any way the notification of the fire department.

401.3.2 Alarm activations. Upon activation of a fire alarm signal, employees or staff shall immediately notify the fire department.

C This section specifically requires immediate notification of the fire department or other emergency response groups when an alarm signal is activated in order to bring emergency forces in the shortest amount of time.

401.3.3 Delayed notification. A person shall not, by verbal or written directive, require any delay in the reporting of a fire to the fire department.

C Emergency plans and procedures must not include a requirement that employees report an alarm to a supervisor or similar person before calling the fire department. There can be no substitute for immediate notification of emergency forces because a quick response is the key to efficient and effective rescue and firefighting.

401.4 Required plan implementation. In the event an unwanted fire is detected in a building or a fire alarm activates, the emergency plan shall be implemented.

C Sections 401.3.1 through 401.3.3 mandate the prompt notification of emergency responders whenever a fire alarm is activated or an actual fire or suspected fire occurs. Since the emergency plan is designed to safely and effectively deal with those circumstances and safeguard the occupants, it must be implemented simultaneously with emergency responder notification as required by this section.

401.5 Making false report. A person shall not give, signal or transmit a false alarm.

C Chapter 2 of the code defines a false alarm as an intentional activation of an alarm or notification of a fire or other emergency when no emergency exists. This would not include a malfunctioning alarm system. False alarms have the potential for causing confusion or panic among occupants of the affected premises, a situation that could lead to property damage, personal injury or death. False alarms also place firefighters and other emergency personnel in potential danger during the unnecessary emergency response. This can jeopardize other lives and property in the community by committing emergency forces to a false situation when they might be needed at an actual emergency elsewhere.

401.6 Emergency evacuation drills. The sounding of a fire alarm signal and the carrying out of an emergency evacuation drill in accordance with the provisions of Section 405 shall be allowed.

C This section specifically allows fire alarm signals to be utilized as part of emergency evacuation drills.

401.7 Unplanned evacuation. Evacuations made necessary by the unplanned activation of a *fire alarm system* or by any other emergency shall not be substituted for a required evacuation drill.

C This section makes it clear that unplanned evacuations will not be applied toward fulfillment of the number of drills required. Evacuation drills are intended to provide for an assessment of the adequacy of an emergency action plan and the response of the building occupants. Occupants may or may not be forewarned of a pending drill depending on the circumstances. Responsible staff prepares for and conducts the drills, a key aspect of which is having monitors in place to assess individual and group performance. An unplanned evacuation does not allow for effective monitoring of performance and, therefore, must not be counted as a required drill.

401.8 Interference with fire department operations. It shall be unlawful to interfere with, attempt to interfere with, conspire to interfere with, obstruct or restrict the mobility of or block the path of travel of a fire department emergency vehicle in any way, or to interfere with, attempt to interfere with, conspire to interfere with, obstruct or hamper any fire department operation.

C A potential hazard when fire departments respond to an emergency is the inability to perform operations because of physical obstructions, restricted mobility or human interference. This section prohibits any type of interference with emergency response operations. The delay of even a few minutes can cause serious property damage, injuries or fatalities.

SECTION 402—DEFINITIONS

402.1 Definitions. The following terms are defined in Chapter 2:

EMERGENCY EVACUATION DRILL.

LOCKDOWN.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2

or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see commentaries, Sections 201.1 through 201.4).

SECTION 403 —EMERGENCY PREPAREDNESS REQUIREMENTS

403.1 General. In addition to the requirements of Section 401, occupancies, uses and outdoor locations shall comply with the emergency preparedness requirements set forth in Sections 403.2 through 403.11.3.3. Where a fire safety and evacuation plan is required by Sections 403.2 through 403.10.6, evacuation drills shall be in accordance with Section 405 and employee training shall be in accordance with Section 406.

C Section 403 contains the occupancy-specific requirements for emergency preparedness. This section primarily notes where a fire safety and evacuation plan is required, but also provides specific requirements as necessary. For instance, seating plans must be provided for Group A occupancies or, in the case of Group I-2 occupancies, specific requirements are provided for staff training. Section 403 also has a section with special requirements to address the need for fire watches and planning for large gatherings (see Section 403.11).

The occupancies addressed by this section were chosen based on the density and location of occupants, the layout of the building or the limitations of the occupants during an emergency. The *Evacuation Planning Guide for Stadiums* published by the US Department of Homeland Security (DHS) provides useful guidance for evacuation planning in large assembly structures. The principles used in that document could also be applied to the other occupancy groups listed in this section. Two additional resources to better understand the evacuation needs of specific types of structures such as tall buildings and airports are the *SFPE Engineering Guide: Fire Safety for Very Tall Buildings and Egress Design Solutions*.

403.2 Group A occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group A occupancies, other than those occupancies used exclusively for purposes of religious worship with an *occupant load* less than 2,000, and for buildings containing both a Group A occupancy and an atrium. Group A occupancies shall comply with Sections 403.2.1 through 403.2.4.

C Group A occupancies are a special concern because of the high density and number of occupants. Additionally, occupants in Group A occupancies are generally not very familiar with the building. Due to lack of losses associated with those attending religious worship services, such occupancies are given an exception to the application of these requirements where the occupant load is less than 2,000 people.

403.2.1 Seating plan. In addition to the requirements of Section 404.2, the fire safety and evacuation plans for assembly occupancies shall include a detailed seating plan, *occupant load* and *occupant load* limit. Deviations from the *approved* plans shall be allowed provided that the *occupant load* limit for the occupancy is not exceeded and the *aisles* and *exit accessways* remain unobstructed.

C Proper planning for an assembly occupancy must consider the number, capacity and physical arrangement of exits. In turn, these factors will dictate how seating may be arranged to prevent obstruction of aisles and exits. The number of seats provided may not exceed what is permitted by the number, arrangement and capacity of exits. Floor area factors are only one element in determining whether the exit capacity is adequate. Additionally, the egress plan must be approved by the fire code official with a copy of the approved plan maintained on the premises for review by employees and inspectors. Many facilities have several approved plans to accommodate various situations and functions. The seating plan selected for any event should reflect the needs of the group and the requirement to keep aisles and exits clear. Deviations from an approved plan may be permitted only if they do not obstruct the complete egress path, including aisles and exits.

403.2.2 Announcements. In theaters, motion picture theaters, auditoriums and similar assembly occupancies in Group A used for noncontinuous programs, an audible announcement shall be made not more than 10 minutes prior to the start of each program to notify the occupants of the location of the *exits* to be used in the event of a fire or other emergency.

Exception: In motion picture theaters, the announcement is allowed to be projected on the screen in a manner *approved* by the *fire code official*.

C Announcements are intended to familiarize occupants with life safety system features potentially needed during a fire. Information is the most valuable commodity during fires and other emergencies; however, it is often difficult, if not impossible, to override the emotion and confusion caused by a fire or other emergency. Studies have generally shown, however, that occupants do not panic as once thought; therefore, it is imperative that occupants receive information necessary for them to make decisions before an emergency occurs. To convey information and motivate an adaptive response to fires or other emergencies, the message must stimulate interest and speak directly to the topic. Many movie theaters currently use “trailers” or “shorts” to market concession items, as well as fire safety. These messages can be especially effective if they are specific and adequately distinguished from other promotions. Any message should reflect the life safety strategy and must point out specific features of the occupancy. Occupants in most theaters and auditoriums are usually expected to leave the building immediately upon notification of a fire using the nearest available exit. This is largely a reflection on the building type and arrangement of exits. In these cases, the locations of all exits must be identified. Raising the house lights along the egress path or modulating aisle lighting at appropriate times during the message can reinforce the message.

This section specifically requires an audible announcement, but the exception for motion picture theaters would allow the message to be visually displayed on the screen as approved. A combination of both an audible and visual message would most likely be the most effective. This section also has a maximum time from the start of the show to provide such announcements. An announcement that comes too early will lose its effectiveness.

403.2.3 Fire watch personnel. Fire watch personnel shall be provided where required by Section 403.11.1.

C This section links to Section 403.11.1, which establishes the authority of the fire code official to require a fire watch as necessary.

403.2.4 Crowd managers. Crowd managers shall be provided where required by Section 403.11.3.

C Assembly occupancies contain many occupants potentially unfamiliar with the building. The larger the number of occupants in a single space, the more critical that crowd managers be available to direct people. Section 403.11.3 requires crowd managers when a gathering exceeds 500 people.

403.3 Group B occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for buildings containing a Group B occupancy where the Group B occupancy has an *occupant load* of 500 or more persons or more than 100 persons above or below the lowest *level of exit discharge* and for buildings having an ambulatory care facility.

C This section requires a fire and evacuation plan where the number of occupants in a Group B occupancy becomes large. The increased risk where such occupants are not located on the level of exit discharge is also addressed. Generally, Group B occupancies are not used for sleeping and occupants are familiar with their surroundings. This section is for Group B occupancies that are not considered ambulatory care facilities. Ambulatory care facilities are also Group B occupancies but are addressed more specifically by Sections 403.3.1 through 403.3.1.2. Group F and M occupancies have the same occupant load and location trigger for fire safety and evacuation plans (see Sections 403.5 and 403.8).

403.3.1 Ambulatory care facilities. Ambulatory care facilities shall comply with the requirements of Sections 401, 403.3.1.1 through 403.3.1.2 and 404 through 406.

C Ambulatory care facilities, similar to Group I-1, Condition 2 and Group I-2 occupancies, use a “defend-in-place” method of occupant protection. Defend-in-place is a widely used approach to protecting occupants who are bedridden, unconscious or otherwise unable to self-preserve in a fire event. The method relies on both active and passive fire protection systems as well as the actions of trained staff and responders. The heavy emphasis on staff action requires a comprehensive fire safety and evacuation plan. Any building containing an ambulatory healthcare occupancy will, by definition, contain occupants who may be incapable of self-preservation. Current IBC requirements for ambulatory care are intended to create a type of defend-in-place environment. Sections 403.3.1.1 through 403.3.1.2 provide requirements for ambulatory care that are more specific than those provided in Sections 404 and 405.

403.3.1.1 Fire safety and evacuation plan. The fire safety and evacuation plan required by Section 404 shall include a description of special staff actions. This shall include procedures for stabilizing care recipients in a *defend-in-place* response, staged evacuation, or full evacuation in conjunction with the entire building if part of a multitenant facility.

C Fire safety and evacuation plans must be developed, reviewed and approved to support this strategy. The reference to “defend-in-place” was added in Section 404.2.2 to recognize the defend-in-place method. This is not a new concept. The IBC and legacy codes have been written to support this concept, and fire safety and evacuation plans must acknowledge this strategy. The term “Defend-in-place” is defined in the IBC. Fire safety plans should describe in the life safety strategy the method of notifying occupants, including the use of a private-mode alarm system as allowed by code. Procedures for dealing with occupants in a defend-in-place strategy should also be described so that it is clear what the staff will be trained on and what the first responders should expect to encounter. Fire evacuation plans are required to describe the special actions of staff, especially staff that must stabilize a care recipient prior to moving. This will be the basis of the staff education and training. It is imperative that the building and fire officials know the size and location of the facility as well as the number of care recipients who are incapable of self-preservation. This information will help the building official determine the proper classification and mitigations required. It will also allow the fire official to preplan the response for a particular building. Any special characteristics of the means of egress, such as path to the adjacent smoke compartment and special locking arrangements, should also be described to aid in verifying code compliance. Practically, these documents will be the basis for staff training as well.

403.3.1.1.1 Fire safety plan. A copy of the fire safety plan shall be maintained at the facility at all times. The plan shall include all of the following in addition to the requirements of Section 404:

1. Locations of care recipients who are rendered incapable of self-preservation.
2. Maximum number of care recipients rendered incapable of self-preservation.

3. Area and extent of each ambulatory care facility.
4. Location of any special locking arrangements.

C This section sets out information required for the fire safety plan specific to ambulatory care facilities. Fire safety plans are required to show the location where incapable care recipients are likely to be, including what portions of the building are included in the ambulatory care facility. In addition, information regarding the maximum number of care recipients and any locking constraints that might affect the horizontal evacuation of care recipients needs to be communicated. All of these will be essential to robust staff training as well as operational planning for first responders.

403.3.1.2 Staff training. Staff shall be periodically instructed and kept informed of their duties and responsibilities under the plan. Records of instruction shall be maintained. Such instruction shall be reviewed by the staff at intervals not exceeding three months. Training of new staff shall be provided promptly on entrance to duty.

Staff shall be instructed in the proper use of portable fire extinguishers and other manual fire suppression equipment.

C In ambulatory care facilities, the actions of the staff are critical to the success of the fire evacuation plan due to the condition of the occupants. Therefore, staff training on a regular basis is necessary. This section requires that the staff reviews the fire evacuation plan and fire safety plan regularly. These plans always need to be available for review. This training specifically requires that staff knows how to use portable fire extinguishers and any other manual fire suppression equipment in order for the fire safety plan to be effective.

403.4 Group E occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group E occupancies and for buildings containing both a Group E occupancy and an atrium. Group E occupancies shall comply with Sections 403.4.1 through 403.4.3.

C Group E occupancies are among the primary occupancies requiring emergency evacuation drills. They are among the few occupancies that require all occupants to participate in evacuation drills. The effectiveness of pre-emergency planning in Group E occupancies has been significant, as evidenced by a remarkable decline in tragic fires in schools over the years. The evacuation skills learned by children in school are often carried with them into adulthood.

403.4.1 First emergency evacuation drill. The first emergency evacuation drill of each school year shall be conducted within 10 days of the beginning of classes.

C Group E occupants vary from year to year as children enter and leave grade levels. In addition, even though a child may be in the same school for a number of years, location within the building and leaders change. It is important, therefore, that the first evacuation drill occurs within the first 10 days of school. This provides students with nearly immediate training and the school and emergency responders with information about where problems exist. Additionally, in more recent years, schools have been immersing students with disabilities into the general classroom. The location and number of these students vary each year. The ability to evacuate them must be assessed early in the school year.

403.4.2 Time of day. Emergency evacuation drills shall be conducted at different hours of the day or evening, during the changing of classes, when the school is at assembly, during the recess or gymnastic periods, or during other times to avoid distinction between drills and actual fires.

C If evacuation drills are done routinely, they will be easily distinguished as drills and not an actual fire. This is potentially dangerous because the behavior patterns will be different and actual preparedness for emergencies will be lessened. Time of day, therefore, should be varied, regardless of whether it is convenient. A fire will not differentiate between a convenient and an inconvenient time.

403.4.3 Assembly points. Outdoor assembly areas shall be designated and shall be located a safe distance from the building being evacuated so as to avoid interference with fire department operations. The assembly areas shall be arranged to keep each class separate to provide accountability of all individuals.

C A key element in safe evacuation is the exit discharge portion of the evacuation route. Once occupants are outside the building, they need to be located far enough away to avoid further hazards. Additionally, there is a potential for people at the assembly point to interfere with the emergency operations of the fire department, which may not be present during an evacuation drill. Locations, therefore, need to be designated such that they avoid hazards and keep the evacuated occupants away from probable paths of emergency response. Also, to simplify accounting for the occupants once outside, the code requires each class to remain together as a group and separate from other classes.

403.5 Group F occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for buildings containing a Group F occupancy where any of the following conditions apply:

1. The Group F occupancy has an *occupant load* of 500 or more persons.
2. The Group F occupancy has an *occupant load* of more than 100 persons above or below the lowest *level of exit discharge*.
3. Group F pallet manufacturing and recycling facilities as required by Section 2810.

C Group F occupancies require a fire safety and evacuation plan under the same occupant load and location conditions as Group B and M occupancies. Essentially, where the occupant load reaches a certain number overall or increases on a level

other than the level of exit discharge, such plans become more critical (see commentary, Section 403.3). Additionally, due to the hazards present at pallet manufacturing and recycling facilities, a fire safety and evacuation plan must be provided and maintained at such facilities.

403.6 Group H occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group H occupancies.

C This section requires that all Group H occupancies have a fire safety and evacuation plan. These occupancies, regardless of size or type, tend to contain significant hazards that make the need for preplanning critical. Semiconductor facilities have more specific requirements found in Section 403.6.1.

403.6.1 Group H-5 occupancies. Group H-5 occupancies shall comply with Sections 403.6.1.1 through 403.6.1.4.

C Group H-5 occupancies are semiconductor fabrication facilities. These types of facilities are unique in that they have a very large allowable area and house a significant amount of hazardous materials. Essentially, the building is divided into fabrication areas, hazardous production materials (HPM) rooms (which are Group H occupancy storage rooms) and networks of service corridors and spaces. The area of the building needs to be large to incorporate all of the operations needed by the semiconductor industry. To facilitate these operations, a special package of requirements and occupancy classification was created. This package is found within the occupancy requirements in the IBC and Chapter 27 of the code. As part of the package, special emergency preparedness and preparation are also required. Sections 403.6.1.1 through 403.6.1.4 provide specific requirements for emergency preparedness and preparation. These requirements are in addition to others pertaining to hazardous materials found in Section 407.

403.6.1.1 Plans and diagrams. In addition to the requirements of Section 404 and Section 407.6, plans and diagrams shall be maintained in *approved* locations indicating the approximate plan for each area; the amount and type of HPM stored, handled and used; locations of shutoff valves for HPM supply piping; emergency telephone locations and locations of *exits*.

C The requirements for plans and diagrams are in addition to those required in Sections 404 and 407. More specifically, the additional details required include the approximate plan for each area of the building. This includes the amount of HPM stored, handled and used in both the fabrication areas and the HPM rooms. Additionally, since it is typical that such facilities tend to pipe HPM throughout the building for efficiency, all shutoff valves must be identified. Finally, exits must be clearly marked. This is specifically necessary because many service corridors for the transport of materials are not considered part of the means of egress. These details will assist both the occupants and the emergency responders.

403.6.1.2 Plan updating. The plans and diagrams required by Sections 404, 403.6.1.1 and 407.6 shall be maintained up-to-date and the *fire code official* and fire department shall be informed of major changes.

C Semiconductor facilities are constantly changing because of the needs of new technology; therefore, the types and amounts of materials and their application are constantly changing. These changes must be accounted for within the plans and diagrams required in Sections 403.6.1.1 and 407; otherwise, the critical information needed by the emergency responders may not be available.

403.6.1.3 Emergency response team. Responsible persons shall be designated as an on-site emergency response team and trained to be liaison personnel for the fire department. These persons shall aid the fire department in preplanning emergency responses, identifying locations where HPM is stored, handled and used, and be familiar with the chemical nature of such material. An adequate number of personnel for each work shift shall be designated.

C This section is similar to Section 407.4 but has some specific requirements unique to semiconductor facilities. More specifically, liaisons familiar with the location of HPM and hazards related to those materials must be on-site. For example, semiconductor facilities make use of silane gas, which is a pyrophoric material—it will instantly ignite when exposed to atmospheric conditions. Knowing the characteristics of the material, where it is located and the quantity used is critical to the responding emergency personnel.

403.6.1.4 Emergency drills. Emergency drills of the on-site emergency response team shall be conducted on a regular basis but not less than once every three months. Records of drills conducted shall be maintained.

C This section requires employees to conduct drills every 3 months to practice specific emergency procedures for the facilities. Again, this requirement is specific to semiconductor facilities because of their unique layout and contents. The emergency responders rely heavily on the on-site actions of the employees because of their intimate knowledge of the site and the complexity of these buildings. Records must be maintained.

403.7 Group I occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group I occupancies. Group I occupancies shall comply with Sections 403.7.1 through 403.7.3.6.

C Group I occupancies are those where occupants are often incapable of self-preservation due to their physical condition or they are physically able but under restraint. Therefore, fire safety and evacuation plans are critical and also have some specific needs unique from other occupancies.

403.7.1 Group I-1 occupancies. Group I-1 occupancies shall comply with Sections 403.7.1.1 through 403.7.1.4.

C A Group I-1 occupancy is one that houses more than 16 individuals who live in a supervised residential care facility on a 24-hour basis. This would include residential board and care homes, congregate care facilities, social rehabilitation facilities, alcohol and drug centers, assisted living facilities and convalescent facilities. Generally, occupants of these facilities are able to respond to an emergency with some assistance from staff. Note that there are two levels of Group I-1 occupancies (Condition 1 and Condition 2). These are further clarified in Section 202 but, essentially, those found in Condition 2 need additional protection due to an increased need for assistance.

403.7.1.1 Fire safety and evacuation plan. The fire safety and evacuation plan required by Section 404 shall include a description of special staff actions. Plans shall include all of the following in addition to the requirements of Section 404:

1. Procedures for full evacuation of care recipients.
2. In Group I-1, Condition 2, procedures for staged evacuation of care recipients through a refuge area in an adjacent *smoke compartment* and then to an exterior assembly point.
3. Shall be amended or revised upon admission of any resident care recipient with unusual needs.

C This section is in addition to the general requirements found in Section 404. It specifies that requirements based on the particular needs of residents must be included as part of the plan. This includes providing procedures for a full evacuation of care recipients. Procedures are required to address the use of refuge areas and smoke compartments for Group I-1, Condition 2 occupancies. To be effective, the plans must be reviewed each time a new care recipient arrives and potentially altered to address any special needs. Group R-4 occupancies are being addressed in a similar manner (see Section 403.9.3.1).

403.7.1.1.1 Fire safety plan. A copy of the fire safety plan shall be maintained at the facility at all times. The plan shall include the following in addition to the requirements of Section 404.2.2:

1. Location and number of care recipient sleeping rooms.
2. Location of special locking arrangements.

C This section requires that Group I-1 occupancies provide the location and number of care recipients sleeping and any special locking arrangements. This assists in search and rescue and understanding how the care recipients may be hampered in their evacuation. This information is critical for preplanning by both staff and the fire department. This applies to both conditions of Group I-1 occupancies, Conditions 1 and 2. The same requirement is provided for Group R-4 occupancies, which are essentially smaller Group I-1 occupancies.

403.7.1.2 Staff training. Staff shall be periodically instructed and kept informed of their duties and responsibilities under the plan. Records of instruction shall be maintained. Such instruction shall be reviewed by staff at intervals not exceeding three months. Training of new staff shall be provided promptly upon entrance to duty.

Staff shall be instructed in the proper use of portable fire extinguishers and other manual fire suppression equipment.

C These types of facilities are normally occupied by care recipients who have the ability to evacuate or relocate with a certain level of assistance from the staff; therefore, staff training is critical. Again, care recipients in Group I-1, Condition 2 occupancies will need more evacuation assistance than in Group I-1, Condition 1 occupancies. Group I-1, Condition 2 occupancies utilize the concept of smoke compartments. Additional training in this regard is necessary for staff. This section requires that training occurs every 3 months because the needs of the occupants may change over time as certain physical or mental conditions progress or new care recipients arrive. New staff must be trained as soon as they begin working as the methods of evacuation will be unique to the facility and the specific care recipients. Staff training also specifically requires training in the proper use of portable fire extinguishers and any other manual fire suppression equipment in order for the fire safety plan to be effective.

403.7.1.3 Resident training. Residents capable of assisting in their own evacuation shall be trained in the proper actions to take in the event of a fire. In Group I-1, Condition 2 occupancies, training shall include evacuation through an adjacent *smoke compartment* and then to an exterior assembly point. The training shall include actions to take if the primary escape route is blocked. Residents shall be trained to assist each other in case of fire to the extent their physical and mental abilities permit them to do so without additional personal risk.

C As noted, Group I-1 occupants are capable of responding to an emergency but will most likely need direction from staff and perhaps physical assistance to ensure the appropriate response. This is more the case with Group I-1, Condition 2 occupancies. Unlike other Group I occupancies, I-1 occupancies rely on the abilities of care recipients to take some level of responsibility for their own evacuation or relocation; therefore, training care recipients in these occupancies is critical. One major element that must be communicated to care recipients is what to do when the main exit route is blocked.

This section also requires the occupants to assist one another as long as a physical or mental condition would not limit their ability to do so.

Since Group I-1, Condition 2 occupancies utilize the concept of smoke compartments, occupants must be trained to evacuate to the adjacent smoke compartment to the refuge area.

403.7.1.4 Drill frequency. In addition to the evacuation drills required in Section 405.3, staff shall participate in drills an additional two times a year on each shift. Twelve drills with all occupants shall be conducted in the first year of operation.

C Occupants of Group I-1 facilities need to be reminded often of evacuation procedures given their sometimes limited abilities to recall procedures, changes in the care recipients' abilities over time and the introduction of new care recipients to the facility. Therefore, this section requires more frequent drills than listed in Table 405.3. Staff on each shift is required to participate in additional drills. Each shift potentially has different staff and also different conditions due to the time of day. Another very specific difference from Table 405.3 is that such facilities are required to have, essentially, monthly drills for the first year. Over time, the occupants and staff will change, but this sets the tone for how the facility operates. This also increases staff awareness of procedures.

403.7.2 Group I-2 occupancies. Group I-2 occupancies shall comply with 403.7.2.1 through 403.7.2.5.

C Group I-2 occupancies as well as ambulatory care facilities and Group I-1, Condition 2 occupancies use a "defend-in-place" method of occupant protection. Defend-in-place is a widely used approach to protect occupants who are bedridden, unconscious or otherwise unable to self-preserve in a fire event. The method relies on both active and passive fire protection systems as well as the actions of trained staff and responders. The heavy emphasis on staff action requires a comprehensive fire safety and evacuation plan. Any building containing an ambulatory healthcare occupancy will, by definition, contain care recipients who may be incapable of self-preservation. Current IBC requirements for Group I-2 occupancies are intended to create a type of defend-in-place environment. Sections 403.7.2.1 through 403.7.2.5 provide requirements for Group I-2 occupancies that are more specific than what is provided in Sections 404 and 405.

403.7.2.1 Fire safety and evacuation plan. The fire safety and evacuation plans required by Section 404 shall include a description of special staff *actions*. Plans shall include all of the following in addition to the requirements of Section 404:

1. Procedures for evacuation for care recipients with needs for containment or restraint and post-evacuation containment, where present.
2. A written plan for maintenance of the *means of egress*.
3. Procedure for a *defend-in-place* strategy.
4. Procedures for a full-floor or building evacuation, where necessary.
5. In Group I-2, Condition 2, amendments or revisions upon admission of any care recipients with unusual needs.

C Staff actions are critical to the success of a fire evacuation plan for Group I-2 occupancies; therefore, procedures need to be documented. In addition to the requirements of Section 404, this section provides four more-specific requirements for the fire evacuation plan. These requirements include how care recipients are to be evacuated and documentation of maintenance plans for means of egress in such facilities. Procedures for a defend-in-place strategy must be documented so that each shift is familiar with that strategy. Defend-in-place procedures depend on the use of refuge areas that need to be maintained. In some cases, the defend-in-place evacuation may need to escalate to full-floor or building evacuation. Such evacuations need to be documented, as well. The fire safety plan needs to be reassessed and potentially changed anytime a care recipient with specific needs is admitted.

403.7.2.2 Fire safety plan. A copy of the fire safety plan shall be maintained at the facility at all times. The plan shall include all of the following in addition to the requirements of Section 404.2.2:

1. Location and number of care recipient sleeping rooms and operating rooms.
2. Location of special locking arrangements.

C Fire safety plans should describe the method of notifying occupants, including the use of a private-mode alarm system as allowed by code. Procedures for dealing with occupants in a defend-in-place strategy should also be described so that it is clear what the staff will be trained on and what the first responders should expect to encounter. It is imperative that the building and fire code officials know the location of the facility as well as the number of care recipients who are incapable of self-preservation. This information will help the building official determine the proper classification and mitigations required. It will also allow the fire official to preplan the response for a particular building. Any special characteristics of special locking arrangements should also be described to aid in verifying code compliance. Practically, these documents will be the basis for staff training as well.

403.7.2.3 Staff training. Staff shall be periodically instructed and kept informed of their duties and responsibilities under the plan. Records of instruction shall be maintained. Such instruction shall be reviewed by staff at intervals not exceeding three months. Training of new staff shall be provided promptly upon entrance to duty.

Staff shall be instructed in the proper use of portable fire extinguishers and other manual fire suppression equipment.

C These types of facilities are normally occupied by care recipients who do not have the ability to evacuate; therefore, staff training is critical. This section requires that training occur every 3 months because the needs of the care recipients change over time as new care recipients are admitted and conditions of care recipients change. New staff must be trained as soon as they begin working as the methods of evacuation will be unique to the facility and the specific care recipients. Staff train-

ing also specifically requires training in the proper use of portable fire extinguishers and any other manual fire suppression equipment in order for the fire safety plan to be effective.

403.7.2.4 Emergency evacuation drills. Emergency evacuation drills shall comply with Section 405.

C Group I-2 occupancies are required to comply with the evacuation drill frequency established in Section 405. Section 405 addresses the specific needs of Group I-2 occupancies, which would not require evacuation to the exterior, and exceptions address the use of coded announcements as an alternative to audible alarms for such drills.

403.7.2.5 Fire loss prevention in operating rooms. Fire protection features and procedures for fire loss prevention in surgical operating rooms shall comply with NFPA 99, Section 16.13.

C Operating rooms are subject to potential fires if flammable materials are exposed to accelerants ignited by cautery or surgical procedures. NFPA 99 is the appropriate standard for systems and procedures to be used in I-2, Condition 2 hospitals to control these hazards to prevent fires in a surgical setting.

403.7.3 Group I-3 occupancies. Group I-3 occupancies shall comply with Sections 403.7.3.1 through 403.7.3.6.

C Group I-3 occupancies are institutional occupancies where the occupants are under restraint. Typically, the occupants are physically able but are restrained from moving freely. Several levels of restraint in such occupancies are described in more detail in Chapter 2 under the definition of “Group I-3 occupancies.” As in Group I-2 occupancies, the only participants in drills will be staff. These requirements are in addition to the requirements of Sections 404 and 405.

403.7.3.1 Fire safety and evacuation plans. The fire safety and evacuation plans required by Section 404 shall include a description of special staff actions. Plans shall include all of the following in addition to the requirements of Section 404:

1. Procedures for evacuation of detainees with needs for containment or restraint and post-evacuation containment, where present.
2. Procedures for a *defend-in-place* strategy.
3. Procedures for a full-floor or building evacuation, where necessary.

C Staff actions are critical to the success of a fire evacuation plan for Group I-3 occupancies; therefore, procedures need to be documented. In addition to the requirements of Section 404, this section provides three specific requirements for the fire safety and evacuation plans. These requirements include how to address detainees that need to be contained or specially restrained during and after the evacuation process. Due to the nature of such facilities, defend-in-place strategies are common and need to be well documented in the fire safety and evacuation plans. Defend-in-place procedures depend on the use of refuge areas that need to be maintained. In some cases, the defend-in-place evacuation may need to escalate to full-floor or building evacuation. Such evacuation plans need to be documented as well.

403.7.3.2 Fire safety plan. A copy of the fire safety plan shall be maintained at the facility at all times. The plan shall include both of the following in addition to the requirements of Section 404.2.2:

1. Location and number of cells.
2. Location of special locking arrangements.

C This section requires that Group I-3 occupancies provide the location and number of cells and any special locking arrangements. This assists in search and rescue and understanding how the detainees may be hampered in their evacuation. This information is critical for preplanning by both staff and the fire department.

403.7.3.3 Staff training. Staff shall be periodically instructed and kept informed of their duties and responsibilities under the plan. Records of instruction shall be maintained. Such instruction shall be reviewed by staff at intervals not exceeding 3 months. Training of new staff shall be provided promptly upon entrance to duty.

Staff shall be instructed in the proper use of portable fire extinguishers and other manual fire suppression equipment.

C Group I-3 facilities are more likely to have incendiary activity; therefore, staff must be trained in the use of various fire protection equipment, including fire extinguishers. Additionally, any new employees must be immediately trained in the use of this equipment. Because of difficulties presented by relocating or evacuating confined occupants, fires should be managed, when practical, to minimize the threat to occupants. Since combustibles are strictly limited in most of these occupancies, accidental fires generally remain small—at least long enough to be manageable. Incendiary fires often pose greater challenges and generally reflect a breakdown in security discipline. Notwithstanding this problem, fire extinguisher training, and even incipient fire brigades, may be especially effective elements of a fire safety plan in restrained-care occupancies. This training is required of new employees before they can begin their official duties. This section requires that staff is periodically instructed on their roles and the overall fire safety and evacuation plans. Records of this training are required to be maintained for reference.

This training specifically requires that staff knows how to use portable fire extinguishers and any other manual fire suppression equipment in order for the fire safety plan to be effective.

403.7.3.4 Staffing. Group I-3 occupancies shall be provided with 24-hour staffing. A staff person shall be within three floors or 300 feet (91 440 mm) horizontal distance of the access door of each resident housing area. In Group I-3 Conditions 3, 4 and 5, as defined

EMERGENCY PLANNING AND PREPAREDNESS

in Chapter 2, the arrangement shall be such that the staff involved can start release of locks necessary for emergency evacuation or rescue and initiate other necessary emergency actions within 2 minutes of an alarm.

Exception: A staff person shall not be required to be within three floors or 300 feet (91 440 mm) horizontal distance of the access door of each resident housing area in areas in which all locks are unlocked remotely and automatically in accordance with Section 408.4 of the *International Building Code*.

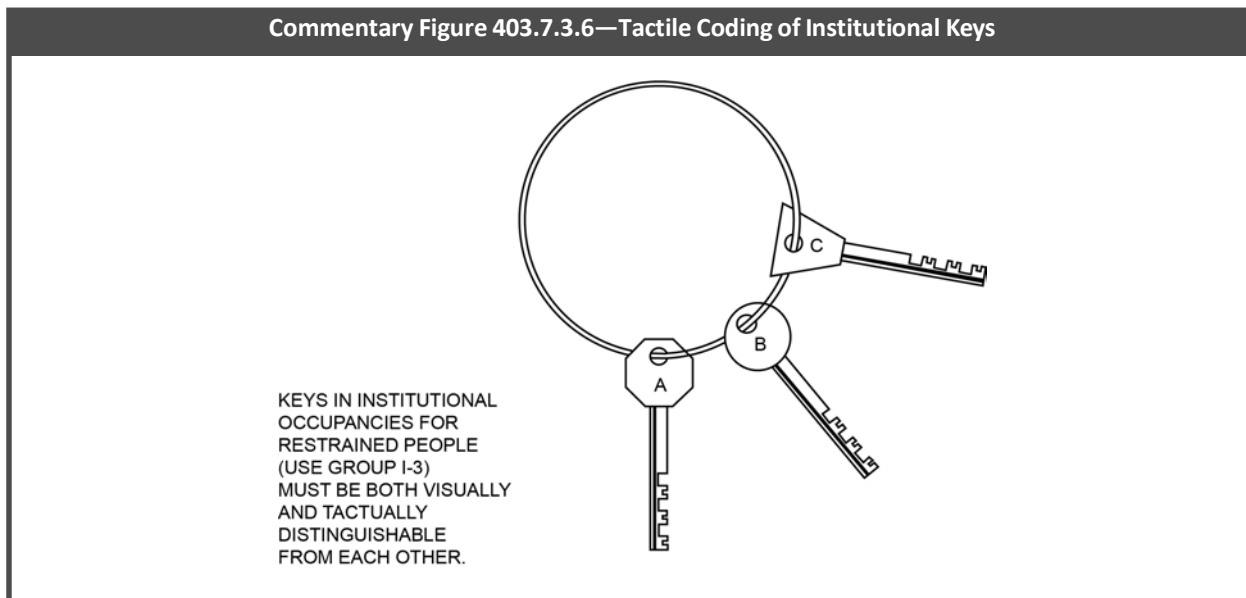
C Group I-3 occupancies place a lot of importance on the actions of staff in emergencies. This section provides specific direction regarding the staff's locations and actions in an emergency. Staff members responsible for initiating the relocation or evacuation of confined occupants must be constantly alert to potential fire hazards and incipient fires. If a fire occurs, 2 minutes will seem like a long time to confined people. Where a remote-release locking system is neither required nor provided, the number of locks requiring manual unlocking should be limited with due regard to staff and confined occupant safety. This may require additional staff to accomplish the unlocking procedure in a timely manner. This section does have a specific exception for systems that utilize a remote locking and unlocking system.

403.7.3.5 Notification. Provisions shall be made for residents in Group I-3 Conditions 3, 4 and 5, as defined in Chapter 2, to readily notify staff of an emergency.

C Group I-3 occupancies under Use Conditions 3, 4 and 5 where the occupants are very limited in their freedom would be considered moderate- and high-security facilities. Because the occupants are so limited and could be located remotely from guards or other staff members as a result of confinement within a compartment, a method is necessary for staff notification of a fire. In open cell blocks, staff members may be within earshot of occupants but generally this requirement necessitates monitors, intercoms or other communication appliances.

403.7.3.6 Keys. Keys necessary for unlocking doors installed in a *means of egress* shall be individually identifiable by both touch and sight.

C Keys must be distinctive from one another so they may be promptly and reliably identified under emergency conditions. Fumbling for the right key can cost valuable seconds, and possibly lives, in the event of a fire (see Commentary Figure 403.7.3.6).



403.8 Group M occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for buildings containing a Group M occupancy where the Group M occupancy has an *occupant load* of 500 or more persons or more than 100 persons above or below the lowest *level of exit discharge* and for buildings containing both a Group M occupancy and an atrium.

C This section requires a fire and evacuation plan where the number of occupants in a Group M occupancy exceeds a specified number, including where such occupants are not located on the level of exit discharge. This is the same requirement as that provided for Group B and F occupancies. However, Group M occupants may not be quite as familiar with their surroundings but the staff in such facilities are available to direct occupants during an emergency. Also, this is an occupancy where people do not sleep (see Sections 403.3 and 403.5).

403.9 Group R occupancies. Group R occupancies shall comply with Sections 403.9.1 through 403.9.3.4.

C This section provides fire safety and evacuation plans specific to Group R-1, R-2 and R-4 occupancies. These are occupancies where people sleep; thus, the need for fire safety and evacuation plans is more critical.

403.9.1 Group R-1 occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group R-1 occupancies. Group R-1 occupancies shall comply with Sections 403.9.1.1 through 403.9.1.3.

C Group R-1 occupancies are residential occupancies that include hotels and boarding houses. These occupancies contain residents that are temporary in nature; therefore, they are more unfamiliar with their surroundings than Group R-2 and R-3 occupants. Sections 403.9.1.1 through 403.9.1.3 provide specific requirements that take into account characteristics of Group R-1 occupancies.

403.9.1.1 Evacuation diagrams. A diagram depicting two evacuation routes shall be posted on or immediately adjacent to every required egress door from each hotel or motel *dwelling unit* or *sleeping unit*.

C This section requires an evacuation plan diagram to be posted in each hotel or motel dwelling unit or sleeping unit. Both the format of the diagram and its location must be approved by the fire code official. This diagram is to display both a primary and a secondary exit route in case a fire or other obstacle blocks the primary route. The plan should be prepared so that the orientation of rooms in relation to the exits is accurately portrayed and easily discernible to the occupant. Often, additional safety information is included on the plan diagram, such as fire alarm box and fire extinguisher locations; however, those added items should not clutter the diagram to the extent that they detract from the clarity of its primary purpose of showing exit routes and exit locations.

403.9.1.2 Emergency duties. Upon discovery of a fire or suspected fire, hotel and motel employees shall perform the following duties:

1. Activate the *fire alarm system*, where provided.
2. Notify the public fire department.
3. Take other action as previously instructed.

C This section contains specific actions employees are to take if a fire occurs. The requirement that they first activate the fire alarm and then call the fire department is intended to avoid a situation in which the employee first investigates or calls security. Immediate notification of the occupants facilitates evacuation or relocation as necessary. Notifying the fire department as early as possible will enable firefighters to reach the building at an earlier stage in the fire.

403.9.1.3 Fire safety and evacuation instructions. Information shall be provided in the fire safety and evacuation plan required by Section 404 to allow guests to decide whether to evacuate to the outside, evacuate to an *area of refuge*, remain in place, or any combination of the three.

C The procedures for isolating occupants from a fire depend on the layout and overall fire protection design of a building. For example, a hotel in a high-rise building may be specifically designed to evacuate in phases; therefore, the occupants need to know the procedures to facilitate a smooth and organized reaction to a fire. The appropriate actions that are available should be communicated. Options may include occupants remaining in their rooms, evacuating or relocating.

403.9.2 Group R-2 occupancies. Group R-2 occupancies shall comply with Sections 403.9.2.1 through 403.9.2.3.

C Group R-2 occupancies are permanent residential occupancies that house multiple occupants in multiple dwelling units. Typically, this includes apartment buildings, dormitories and other related residential occupancies. Generally, the occupants tend to be familiar with their surroundings but may be sleeping when an emergency occurs. Sections 403.9.2.1 through 403.9.2.3 provide specific requirements in addition to the general requirements of Chapter 4. This section deals with Group R-2 occupancies generally and, more specifically, with Group R-2 college and university buildings.

403.9.2.1 College and university buildings. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group R-2 college and university buildings. Group R-2 college and university buildings shall comply with Sections 403.9.2.1.1 and 403.9.2.1.2.

C This section establishes the applicability of certain provisions to college and university residence halls and dormitories classified in Group R-2. Fire represents a significant risk to life and property in dormitory occupancies, particularly at colleges and universities. The large number of young people living in close proximity to one another creates the potential for a relatively small fire to have serious and possibly fatal consequences. Applying these requirements in conjunction with the drills required by Table 405.3 will enhance the likelihood of occupants being familiar with exiting buildings under emergency conditions at varying times throughout the day.

403.9.2.1.1 First emergency evacuation drill. The first emergency evacuation drill of each school year shall be conducted within 10 days of the beginning of classes.

C The residents of Group R-2 college and dormitory occupancies typically change each year. It is important, therefore, that the first evacuation drill occurs within the first 10 days of school.

403.9.2.1.2 Time of day. Emergency evacuation drills shall be conducted at different hours of the day or evening, during the changing of classes, when school is at assembly, during recess or gymnastic periods or during other times to avoid distinction between drills and actual fires. One required drill shall be held during hours after sunset or before sunrise.

C If evacuation drills are done routinely, they will be easily distinguished as drills and not an actual fire. This is potentially dangerous because the behavior patterns will be different and actual preparedness for emergencies will be lessened. Time of day, therefore, should be varied, regardless of whether it is convenient. A fire will not differentiate between a convenient and an inconvenient time. At least one of the drills required by Table 405.3 must be conducted between sunset and sunrise so that residents can experience nighttime evacuation.

403.9.2.2 Emergency guide. Fire emergency guides shall be provided for Group R-2 occupancies. Guide contents, maintenance and distribution shall comply with Sections 403.9.2.2.1 through 403.9.2.2.3.

C The next few sections lay out the requirements for emergency guides for Group R-2 occupancies. This includes what is to be in the guide, obtaining approval from the fire code official and distribution requirements. The guide in general will better inform the residents and provide preplanning tools for the fire department.

403.9.2.2.1 Guide contents. A fire emergency guide shall describe the location, function and use of fire protection equipment and appliances available for use by residents, including fire alarm systems, smoke alarms and portable fire extinguishers. Guides shall include an emergency evacuation plan for each *dwelling unit*.

C The guide must contain the intended evacuation plan for each unit and information about the various fire protection features provided to occupants. Providing this information to residents increases the likelihood of a proper response, which in turn increases resident safety and makes the fire department's job a little easier when responding to a scene.

403.9.2.2.2 Emergency guide maintenance. Emergency guides shall be reviewed and *approved* by the *fire code official*.

C This section makes it clear that the fire code official must sign off on the emergency guide contents. This will make fire department response consistent with the features of the building.

403.9.2.2.3 Emergency guide distribution. A copy of the emergency guide shall be given to each tenant prior to initial occupancy.

C This section requires that a guide illustrating the fire safety features of the building be provided to each dwelling unit. There is no purpose in having a guide to assist residents if there is no requirement that it be distributed. This must occur before they become occupants.

403.9.2.3 Evacuation diagrams for dormitories. A diagram depicting two evacuation routes shall be posted on or immediately adjacent to every required egress door from each dormitory *sleeping unit* or *dwelling unit*. Evacuation diagrams shall be reviewed and updated as needed to maintain accuracy.

C The same concept is used for Group R-1 occupancies; this simply increases the information available to occupants. Group R-2 dormitory occupants are typically more transient than most occupants of Group R-2 occupancies (see commentary, Section 403.9.1.1).

403.9.3 Group R-4 occupancies. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for Group R-4 occupancies. Group R-4 occupancies shall comply with Sections 403.9.3.1 through 403.9.3.4.

C A Group R-4 occupancy is a residential care/assisted living facility for more than five but not more than 16 residents. The occupants are similar to those found in a Group I-1 occupancy; therefore, they are capable of self-preservation but may have mental or physical conditions that could impede their reactions.

403.9.3.1 Fire safety and evacuation plan. The fire safety and evacuation plan required by Section 404 shall include a description of special staff actions. Plans shall include procedures necessary for full evacuation of care recipients, and shall be amended or revised upon admission of care recipients with unusual needs.

C This section is in addition to the general requirements found in Section 404. It specifies that requirements based on the particular needs of care recipients must be included as part of the plan and that a full-building evacuation needs to be part of the evacuation plan. These plans must be reviewed each time a new care recipient arrives to assess any special features that need to be included in the plan in order to address those particular needs. This is the same requirement as provided for Group I-1 occupancies.

403.9.3.1.1 Fire safety plan. A copy of the fire safety plan shall be maintained at the facility at all times. The plan shall include the following in addition to the requirements of Section 404.2.2:

1. Location and number of care recipient sleeping rooms.
2. Location of special locking arrangements.

C This section requires that fire safety plans in Group R-4 occupancies provide the location and number of care recipients sleeping and any special locking arrangements. This assists in search and rescue and understanding how the care recipients may be hampered in their evacuation. This information is critical for preplanning by both staff and the fire department. This is the same requirement as provided for Group I-1 occupancies (see Section 403.7.1.1.1).

403.9.3.2 Staff training. Staff shall be periodically instructed and kept informed of their duties and responsibilities under the plan. Records of instruction shall be maintained. Such instruction shall be reviewed by staff at intervals not exceeding three months. Training of new staff shall be provided promptly upon entrance to duty.

Staff shall be instructed in the proper use of portable fire extinguishers and other manual fire suppression equipment.

C These types of facilities are normally occupied by people who have the ability to evacuate or relocate with a certain level of assistance from the staff; therefore, employee training is critical. Residents of Group R-4, Condition 2 occupancies will need more assistance in evacuation than in Group R-4, Condition 1. This section requires that training occur every 3 months because the needs of the occupants may change over time as certain physical or mental conditions progress or new occupants arrive. New staff must be trained as soon as they begin working as the methods of evacuation will be unique to the facility and the specific care recipients. Staff training also specifically requires training in the proper use of portable fire extinguishers and any other manual fire suppression equipment in order for the fire safety plan to be effective.

403.9.3.3 Resident training. Residents capable of assisting in their own evacuation shall be trained in the proper actions to take in the event of a fire. The training shall include actions to take if the primary escape route is blocked. Residents shall be trained to assist each other in case of fire to the extent their physical and mental abilities permit them to do so without additional personal risk.

C As noted, Group R-4 occupants are capable of responding to an emergency, but most likely will need direction from staff and perhaps physical assistance. This is more the case with Group R-4, Condition 2 occupancies. Group R-4 occupancies, similar to Group I-1 occupancies, rely on the abilities of care recipients to take some level of responsibility for their own evacuation or relocation; therefore, training residents in these occupancies is critical. One major element that must be communicated to care recipients is what to do when the main exit route is blocked.

This section also requires the care recipients to assist one another as long as a physical or mental condition would not limit their ability to do so.

403.9.3.4 Drill frequency. In addition to the evacuation drills required in Section 405.3, staff shall participate in drills an additional two times a year on each shift. Twelve drills with all occupants shall be conducted in the first year of operation.

C Occupants of Group R-4 facilities need to be reminded often of evacuation procedures given their sometimes limited capabilities to recall procedures, changes in their abilities over time and the introduction of new residents to the facility. Therefore, this section requires more frequent drills than Table 405.3. Employees on each shift are required to participate in additional drills. Each shift potentially has different staff and also different conditions due to the time of day. Another very specific difference from Table 405.3 is that such facilities are required to have, essentially, monthly drills for the first year. Over time the occupants and staff will change, but this sets the tone for how the facility operates. This also increases staff awareness of procedures.

403.10 Special uses. Special uses shall be in accordance with Sections 403.10.1 through 403.10.6.1.

C The next several sections are focused on special uses that require fire safety and evacuation plans, including:

- Malls and mall buildings (covered and open).
- High-rise buildings.
- Underground buildings.
- Occupant evacuation elevators.
- Buildings containing High piled combustible storage.

403.10.1 Covered and open mall buildings. Covered and open mall buildings shall comply with the requirements of Sections 403.10.1.1 through 403.10.1.6.

C A covered mall building is a special use as described in IBC Section 402 and as defined in Chapter 2 of the code. Generally, a covered mall building is a single building housing multiple occupancies, including but not limited to retail, assembly, drinking, dining and entertainment in which two or more tenants have a main entrance into a mall area. A mall area is also defined in IBC Chapter 2 as “a roofed or covered mall building that serves as access for two or more tenants and does not exceed three levels that are open to each other”; therefore, it presents some unique issues concerning fire department response to an emergency.

An open mall is an uncovered common pedestrian walk that is open to the sky above and to tenant spaces within the open mall building, and typically connects to the anchor buildings. Unless noted otherwise, open malls must comply with all IBC provisions. The open mall building includes all of the buildings wherein two or more tenants have a main entrance into one or more open malls. Because open malls are characterized by there not being a roof connecting one side of the pedestrian mall to the other, the open mall “building” may actually be a collection of separate buildings that all rely on a shared pedestrian concourse for egress. Similar to the covered mall building, the open mall “building” does not include the anchor buildings. Unless noted otherwise, open mall buildings have to comply with all provisions for covered mall buildings.

The requirements of Sections 403.10.1.1 through 403.10.1.6 are related primarily to the complexity of the building and provide appropriate information to emergency responders so they can more effectively respond to a fire or other emergency.

403.10.1.1 Malls and mall buildings exceeding 50,000 square feet. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for covered malls exceeding 50,000 square feet (4645 m²) in aggregate floor area and for open mall buildings exceeding 50,000 square feet (4645 m²) in aggregate area within the perimeter line.

C The requirement for fire safety and evacuation plans for both covered malls and open malls is applicable only when such malls become larger. Smaller malls simplify the evacuation process and detailed information such as a lease plan is not required.

403.10.1.2 Lease plan. In addition to the requirements of Section 404.2.2, a lease plan that includes the following information shall be prepared for each covered and open mall building:

1. Each occupancy, including identification of tenant.
2. *Exits* from each tenant space.
3. Fire protection features, including the following:
 - 3.1. Fire department connections.
 - 3.2. *Fire command center*.
 - 3.3. Smoke management system controls.
 - 3.4. Elevators, elevator machine rooms and controls.
 - 3.5. Hose valve outlets.
 - 3.6. Sprinkler and standpipe control valves.
 - 3.7. Areas protected with *automatic sprinkler systems* and *automatic fire-extinguishing systems*.
 - 3.8. Automatic fire detector zones.
 - 3.9. *Fire barriers*.

C Item 1 assists the emergency responders by requiring detailed documentation regarding the identification of each tenant, location and occupancy. This will let them locate the highest density of occupants and the types of hazards that may be anticipated.

Item 2 is the identification of exits. This will assist emergency responders in the identification of necessary access routes and how they may interact with the exits.

Item 3 is a report of the available fire protection features. These features, such as identification of the fire detector zones, will help responders quickly assess where a fire is located. If the mall has a smoke control system, access to the controls may be necessary; therefore, the location of those controls is critical. This information gives the fire department a general feel for how the building is intended to perform during a fire. This information is valuable in the sense that they will have more information to promote effective use of the fire protection features installed. If little information is provided, the fire department could actually disrupt the essential activation of a system, such as smoke control.

403.10.1.3 Lease plan approval. The lease plan shall be submitted to the *fire code official* for approval, and shall be maintained on-site for immediate reference by responding fire service personnel.

C The lease plan must be approved. This allows the fire department to determine whether all necessary information, from the perspective of the responders, is addressed. Also, this section requires that the lease plan is available on-site for use by the emergency responders.

403.10.1.4 Lease plan revisions. The lease plans shall be revised annually or as often as necessary to keep them current. Modifications or changes in tenants or occupancies shall not be made without prior approval of the *fire code official* and building official.

C This section provides the authority to require a review of the lease plan at least once each year. In addition, the fire code official has the authority to ask for more frequent reviews of the plan.

If a change occurs in the building, the lease plan may no longer be valid; therefore, this section requires that no changes to any tenant space be made without approval and review by the fire code official. These changes would have to be documented in the lease plan.

403.10.1.5 Tenant identification. Tenant identification shall be provided for secondary *exits* from occupied tenant spaces that lead to an *exit corridor* or directly to the exterior of the building. Tenant identification shall be posted on the exterior side of the *exit* or *exit access door* and shall identify the business name and address using plainly legible letters and numbers that contrast with their background.

Exception: Tenant identification is not required for anchor stores.

C Identifying secondary exits from tenant spaces that enter into an exit passageway or lead directly outside is a critical need for emergency responders. This identification needs to contain the business name and address. Having multiple tenants within a building makes this identification necessary. Anchor stores do not need these labels, since they are fairly recognizable without them.

403.10.1.6 Unoccupied tenant spaces. The fire safety and evacuation plan shall provide for compliance with the requirements for unoccupied tenant spaces in Section 311.

C This section addresses the hazards posed by a tenant space that is not in use or that is not under the supervision of employees. These spaces are more likely to be targeted by vandalism and possibly incendiary activity. Generally, a fire can grow unnoticed in such spaces as a result of the lack of supervision or activity in the space. This section simply references Section 311.6, where requirements dealing with vacant premises are found. To reduce the risks in unsupervised tenant spaces, several requirements in Section 311.6 focus on reducing the fire ignition, growth and spread potential by limiting combustibles in the space, securing the space through the use of locks and installing fire separations constructed of 1/2-inch-thick (12.7 mm) gypsum or similar materials. Note that mall buildings are required to be sprinklered throughout and that the systems in the tenant spaces must be independent of the mall area.

403.10.2 High-rise buildings. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for high-rise buildings.

C High-rise buildings can involve many different uses, such as business, assembly and residential. The types of occupancies on their own may or may not have specific requirements for fire safety and evacuation plans, but due to the building being tall, evacuation and fire safety become more complex. Such buildings tend to use phased evacuation but may have other unique evacuation requirements such as the need for full-building evacuation or assisting those who cannot use stairways. The types of evacuation strategies for high rises vary. More detailed discussion on these strategies can be found in the *SFPE Engineering Guide: Fire Safety for Very Tall Buildings* (ICC and SFPE 2021) and *Egress Design Solutions* (Tubbs and Meacham 2005).

403.10.3 Underground buildings. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for underground buildings.

C Like high-rise buildings, underground buildings can contain a variety of uses. Those uses may not, on their own, require unique fire safety and evacuation plans, but since the building is located below grade, evacuation and firefighting become more complex. Therefore, it is important that a fire safety and evacuation plan be required in accordance with Section 404. Note that Item 4 in Section 404.2.1 specifically addresses those unable to use the stairways.

403.10.4 Buildings using occupant evacuation elevators. In buildings using occupant evacuation elevators in accordance with Section 3008 of the *International Building Code*, the fire safety and evacuation plan and the training required by Sections 404 and 406, respectively, shall incorporate specific procedures for the occupants using such elevators.

C This is not a specific use but instead is an evacuation strategy that is very different than what the general public has been taught about the use of elevators during fires. The successful use of such elevators depends greatly on preplanning and training as to how they are to be used by building occupants. This section requires that where these elevators are used, specific procedures be worked into the fire safety and evacuation plans and into employee training. Note that Item 3 in Section 404.2.1 specifically requires these procedures to be included in the fire evacuation plans. It should be noted that this ties in with the requirements for high-rise buildings and how they are intended to be evacuated. As with high-rise buildings, a good resource for extensive discussion on occupant evacuation and evacuation using elevators can be found in the *SFPE Engineering Guide: Fire Safety for Very Tall Buildings* (ICC and SFPE 2021) and *Egress Design Solutions* (Tubbs and Meacham 2005).

403.10.5 Buildings with high-piled storage. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared for buildings with *high-piled combustible storage* in any of the following situations:

1. The *high-piled storage area* exceeds 500,000 square feet (46 450 m²) for Class I-IV commodities.
2. The *high-piled storage area* exceeds 300,000 square feet (27 870 m²) for high-hazard commodities.
3. The *high-piled storage* is located in a Group H occupancy.
4. The *high-piled storage* is located in a Group F occupancy with an *occupant load* of 500 or more persons or more than 100 persons above or below the lowest *level of exit discharge*.
5. The *high-piled storage* is located in a Group M occupancy with an *occupant load* of 500 or more persons or more than 100 persons above or below the lowest *level of exit discharge*.
6. Where required by the *fire code official* for other *high-piled storage areas*.

C This section correlates the requirement for a fire safety and evacuation plan for high-piled storage with the current requirements in Chapter 32. High-piled storage is a content-based use that has a large effect on the fire safety and evacuation aspects of a building. Chapter 32 requires that fire safety and evacuation plans be in compliance with this section. Prior to the 2018 code, those compliance details were addressed within Chapter 32. Placement in Chapter 4 is more consistent with other occupancies and uses being addressed in this chapter.

There are six thresholds as to when a fire safety and evacuation plan is required for high-piled combustible storage.

In Table 3206.2, high-piled storage areas of Class I-IV commodities over 500,000 square feet currently require additional fire protection measures. Therefore, the situation described in Item 1 is an appropriate time to request a fire safety and evacuation plan. Likewise, Item 2 addresses the fact that in Table 3206.2, high-piled storage areas of high-hazard

commodities over 300,000 square feet currently require additional fire protection measures. Therefore, this situation is also an appropriate time to ask for a fire safety and evacuation plan.

Items 3–5 are consistent with Sections 403.5, 403.6 and 403.8.

Item 6 was previously the primary trigger for requiring a fire safety and evacuation plan specific to high-piled combustible storage. Now this is simply one of the criteria and retains that authority where it is felt necessary.

403.10.6 Lithium-ion and lithium metal batteries. An *approved* fire safety and evacuation plan in accordance with Section 404 shall be prepared and maintained for occupancies that involve activities for the research and development, testing, manufacturing, handling or storage of lithium-ion batteries or lithium metal batteries, or the repair or servicing of vehicles powered by lithium-ion batteries or lithium metal batteries.

Exceptions: A fire safety and evacuation plan is not required for the storage or merchandizing of any of the following:

1. New or refurbished batteries installed for use in the equipment or vehicles they are designed to power.
2. New or refurbished batteries packed for use with the equipment or vehicles they are designed to power for merchandizing purposes.
3. New or refurbished lithium-ion batteries rated at not more than 300 watt-hours and lithium metal batteries containing not more than 25 grams of lithium metal in their original retail packaging.
4. The storage, repair and charging activities in detached one- and two-family *dwelling units* and townhouses, provided that such devices are for personal use.
5. The storage, repair and charging activities associated with personal use in *sleeping units* and *dwelling units* of Group R-1 and R-2 occupancies.

C This section addresses the hazards of lithium-ion or lithium metal batteries. The provisions recognize the fire hazards associated with all phases of lithium-ion or lithium metal battery life cycles: research and development, testing, manufacturing, handling or storage, and vehicle repair or servicing.

Lithium energy storage sources are used in many industries, whether for basic power sources for appliances and vehicles or for standby or alternative power. Statistics have not been fully analyzed for fire frequency, but according to the US Consumer Product Safety Commission in 2020, an average of one fire every 2 days occurred somewhere in the country. Every year, 1.5 percent of lithium-ion batteries are linked to overheating, explosion or fire incidents. It is important to also understand the applications for such batteries are extremely varied and some applications are far less hazardous than others.

Lithium-ion battery fires happen for a variety of reasons, such as physical damage (e.g., the battery is penetrated or crushed or exposed to water), electrical damage (e.g., overcharging or using charging equipment not designed for the battery), exposure to extreme temperatures, and product defects. Thirty percent to 40 percent of lithium-ion battery fires are caused by external factors, such as short-circuiting, overcharging and overdischarging.

The exceptions describe locations where storage and merchandising uses (usually Group S-1 or M) do not have to have fire safety and evacuation plans based solely on the presence of lithium-ion or lithium metal batteries. See Section 403.8 (Group M occupancies) for fire and evacuation plans based on occupant load.

The exceptions also omit fire safety and evacuation plans for one- and two-family dwellings, townhouses, and Group R-1 or R-2 sleeping units and dwelling units used for personal battery storage, repair or charging.

403.10.6.1 Mitigation planning. The *approved* fire safety and evacuation plan shall include thermal runaway event mitigation measures. These measures shall include activities undertaken to prevent thermal runaway, early detection of a thermal runaway event and mitigation measures to be undertaken to limit the size and impact of the event on occupants and the facility.

C The fire safety and evacuation plan must take into consideration mitigation planning for a thermal runaway event involving lithium-ion and lithium metal batteries. Early mitigation is a key to preventing events and controlling the size of the event should one occur.

403.11 Special requirements for public safety. Special requirements for public safety shall be in accordance with Sections 403.11.1 through 403.11.3.3.

C The following requirements authorize a jurisdiction to require fire watches, address large public assemblies and provide specific requirements for crowd managers.

403.11.1 Fire watch personnel. Where, in the opinion of the *fire code official*, it is essential for public safety in a place of assembly or any other place where people congregate, because of the number of persons, or the nature of the performance, exhibition, display, contest or activity, the *owner*, agent or lessee shall provide one or more fire watch personnel, as required and *approved*. Fire watch personnel shall comply with Sections 403.11.1.1 and 403.11.1.2.

C Even though Chapter 31 requires standby personnel in tents and membrane structures because of the inherently higher life safety risks associated with such occupancies, this section gives the fire code official the authority to require fire watch personnel in indoor or outdoor venues where people congregate where the number of persons or the nature of the performance, exhibition, display, contest or activity is such that the presence of fire watch personnel are essential to public safety (see commentary, Section 202, definition of “Fire watch”).

403.11.1.1 Duty times. Fire watch personnel shall remain on duty while places requiring a fire watch are open to the public, or when an activity requiring a fire watch is being conducted.

C This section establishes the key times a fire watch should be conducted. Once it is established that a fire watch is necessary, it is critical that it be conducted while the public is present. Also, it may be necessary for the fire watch to occur based on a specific hazard or activity even while not open to the public.

403.11.1.2 Duties. On-duty fire watch personnel shall have the following responsibilities:

1. Keep diligent watch for fires, obstructions to *means of egress* and other hazards.
2. Take prompt measures for remediation of hazards and extinguishment of fires that occur.
3. Take prompt measures to assist in the evacuation of the public from the structures.

C Fire watch personnel provide temporary fire safety where there are potential fire and life safety hazards that could affect large numbers of assembled people. Such personnel do not simply watch for a fire but are also present to prevent fire by identifying and controlling fire hazards, monitoring and maintaining the availability of the means of egress, and taking initial action to suppress a fire should one occur. Fire watch personnel also provide a method of notifying the fire department if a fire should occur.

403.11.2 Public safety plan for gatherings. Where the *fire code official* determines that an indoor or outdoor gathering of persons has an adverse impact on public safety through diminished access to buildings, structures, fire hydrants and fire apparatus access roads or where such gatherings adversely affect public safety services of any kind, the *fire code official* shall have the authority to order the development of or prescribe a public safety plan that provides an *approved* level of public safety and addresses the following items:

1. Emergency vehicle ingress and egress.
2. Fire protection.
3. Emergency egress or escape routes.
4. Emergency medical services.
5. Public assembly areas.
6. The directing of both attendees and vehicles, including the parking of vehicles.
7. Vendor and food concession distribution.
8. The need for the presence of law enforcement.
9. The need for fire and emergency medical services personnel.
10. The need for a weather monitoring person.

C This section is important because it grants the fire code official the authority to require the development of or to prescribe a specific plan for large gatherings. Such gatherings could include outdoor festivals, demonstrations or receptions. If such assemblies include the use of tents and canopies, Chapter 31 would also apply.

Again, the primary aim of this section is to address the fact that these large gatherings may hamper the ability of the fire department and other emergency responders to access and protect buildings and building occupants.

Further, this section provides some specific issues to be addressed, including: the direction of traffic; vendor and food concession distributors; and the need for law enforcement and medical services.

403.11.3 Crowd managers. Where facilities or events involve a gathering of more than 500 people, crowd managers shall be provided in accordance with Sections 403.11.3.1 through 403.11.3.3.

C This section is similar to the requirement for standby personnel for tents in Section 3108.16.2. Also note that Section 3107.4.3 addresses outdoor assembly events and requires crowd managers, meeting the requirements of this section where crowds exceed 1,000 people. There is an exception for outdoor gatherings less than 1,000 people in Section 403.11.3.1, which is consistent with Section 3107.4.3 and would not require compliance with this section. Large assemblies of people create the need for crowd management due to the increased potential for panic and fear in emergency situations. It is intended that crowd managers can be personnel already assigned and employed by the facility, provided that they are trained as crowd managers to fulfill this requirement. At the time of an emergency, the trained crowd managers would take on additional responsibilities to control and direct the audience or attendees in a safe manner.

403.11.3.1 Number of crowd managers. Not fewer than two trained crowd managers, and not fewer than one trained crowd manager for each 250 persons or portion thereof, shall be provided for the gathering.

Exceptions:

1. Outdoor events with fewer than 1,000 persons in attendance shall not require crowd managers.
2. Assembly occupancies used exclusively for religious worship with an *occupant load* not exceeding 1,000 shall not require crowd managers.

3. The number of crowd managers shall be reduced where, in the opinion of the *fire code official*, the fire protection provided by the facility and the nature of the event warrant a reduction.

C The minimum number of crowd managers would be two, based on the minimum requirements of Section 403.11.3. Additionally, this section would also require an additional crowd manager for any portion of 250 people after that. For instance, if you had an event with 530 people, you would need three crowd managers. Exception 1 addresses consistency with the requirements of Section 3107.4.3 for outdoor events that would not require crowd managers until the number of people in attendance exceeds 1,000. This is related to the fact that outdoor events eliminate many hazards since people can more freely leave such areas. Exception 2 is for assembly occupancies with 1,000 or fewer occupants where religious worship occurs, as such facilities have low risk of fire and life safety concerns. Because the fire and life safety profile of a facility is improved where an automatic sprinkler system is installed throughout, Exception 3 states that the fire code official is authorized to reduce the crowd manager-to-occupant ratio for events on a case-by-case basis (see commentary, Section 3108.16).

403.11.3.2 Training. Training for crowd managers shall be *approved*.

C Training is critical for crowd managers to be effective. This section provides the mechanism to ensure such training occurs. There are various crowd manager training courses available.

403.11.3.3 Duties. The duties of crowd managers shall include, but not be limited to:

1. Conduct an inspection of the area of responsibility and identify and address any egress barriers.
2. Conduct an inspection of the area of responsibility to identify and mitigate any fire hazards.
3. Verify compliance with all permit conditions, including those governing pyrotechnics and other special effects.
4. Direct and assist the event attendees in evacuation during an emergency.
5. Assist emergency response personnel where requested.
6. Other duties required by the *fire code official*.
7. Other duties as specified in the fire safety plan.

C The exact duties and responsibilities of individuals employed as crowd managers are not defined here other than the requirement that they be trained and present. They could serve as ushers, tour guides, service supervisors for table seating or in some other capacity related to making sure occupants are moved to or from assigned places in an orderly way. The key to the success of this section is that crowd managers must be trained in crowd management procedures appropriate to the activity being carried on in the facility and they must be present in the required numbers. Training of personnel and the duties assigned to them would have to be approved by the fire code official. See the commentary to Section 3108.16 for further information on crowd managers.

SECTION 404—FIRE SAFETY, EVACUATION AND LOCKDOWN PLANS

404.1 General. Where required by Section 403 or other sections of this code, fire safety, evacuation and lockdown plans shall comply with Sections 404.2 through 404.4.1.

C This section simply states that all fire safety, evacuation and lockdown plans must comply with Section 404. Section 403 outlines, for many different occupancies and uses, when this section would apply. Other sections of the code may also require compliance with Sections 404.2 through 404.4.1. Section 403 provides specific requirements that are intended to be addressed along with the requirements in Section 404.

404.2 Contents. Fire safety, evacuation and lockdown plan contents shall be in accordance with Sections 404.2.1 through 404.2.3.2.

C The three primary plans addressed by Section 404.3 are a fire evacuation plan, a fire safety plan and a lockdown plan. The fire evacuation plan focuses primarily on the procedures for the evacuation of the occupants in an emergency. The fire safety plan focuses on the overall understanding of the fire protection package of the building as it pertains to the layout of the building, the contents of the building, the means of egress system, the fire hazards and the identification of key contacts during an emergency. Lockdown plans are permitted with approval by the fire code official and address security concerns while also maintaining the fire and life safety of the occupants.

404.2.1 Fire evacuation plans. Fire evacuation plans shall include the following:

1. Emergency egress or escape routes and whether evacuation of the building is to be complete by selected floors or areas only or with a *defend-in-place* response.
2. Procedures for employees who must remain to operate critical equipment before evacuating.
3. Procedures for the use of elevators to evacuate the building where occupant evacuation elevators complying with Section 3008 of the *International Building Code* are provided.
4. Procedures for assisted rescue for persons unable to use the general *means of egress* unassisted.
5. Procedures for accounting for employees and occupants after evacuation has been completed.
6. Identification and assignment of personnel responsible for rescue or emergency medical aid.

7. The preferred and any alternative means of notifying occupants of a fire or emergency.
8. The preferred and any alternative means of reporting fires and other emergencies to the fire department or designated emergency response organization.
9. Identification and assignment of personnel who can be contacted for further information or explanation of duties under the plan.
10. A description of the emergency voice/alarm communication system alert tone and preprogrammed voice messages, where provided.

C The primary focus of evacuation plans is to prepare for and define roles for evacuation and relocation of occupants during an emergency. The fire evacuation plan is important for both emergency responders and building or facility occupants. It focuses the occupants' activities on facilitating a smoother evacuation or relocation process and provides the fire department with critical information on the building and the location of the occupants. Keep in mind that these requirements apply to all occupancies listed in Section 404.2. The occupancy- and use-specific requirements are located within Section 403; therefore, the requirements listed here are general and will vary based on many factors, such as the occupants' mobility and familiarity with the building.

Item 1 requires that specific escape routes be defined. This is important because the building is generally designed to facilitate a particular pattern of evacuation or relocation in an emergency. For instance, as noted earlier, a high-rise building will most likely be evacuated in phases. If floors begin evacuating before intended, the evacuation of the occupants in the fire area may be delayed. Also, if everyone tries to use the same exits in a facility, such as a multiplex theater, evacuation of the building will be delayed. As stated earlier, studies have shown that people tend to exit the way they enter a building. Note that the code sometimes requires a certain level of redundancy to account for occupants using the same exits. For example, the IBC requires that the main exit of multiplex theaters be sized for at least half of the occupants even though plenty of egress width may be available elsewhere in the building. The more coordinated the plan, the more evenly the exits will be used.

Item 2 requires that specific procedures for evacuation be provided to those employees who must operate critical equipment before evacuation. These procedures are necessary to ensure a clear understanding to the occupant when evacuation is critical and the operation should be abandoned.

Item 3 requires that specific procedures be provided when occupant evacuation elevators are used. This is critical as each building may have unique aspects regarding the use of such elevators and the types of evacuations they may undertake.

Item 4 provides for the inclusion of procedures relative to those occupants unable to evacuate using the general egress features who, therefore, may need assisted rescue. It is important that there be adequate information and procedures established and available to staff and firefighters so that assistance can be offered to anyone who needs help using the general means of egress as quickly as possible.

Item 5 simply states that a plan be developed to account for all occupants after evacuation or relocation. This is important not only to the occupants but also to the emergency responders to assess their actions when arriving at the scene.

Item 6 has two roles. First, it provides a designated person for occupants to look to for assistance in an emergency. This will reduce the stress of the situation. Second, when the emergency responders arrive they will have a specific contact to help them assess the situation. These contacts can also be beneficial to emergency responders when preplanning their response to that specific facility.

Item 7 requires that the notification to the occupants of the emergency be standardized. The approach will vary based on occupancy and use. For instance, all occupants in a multiplex theater would be notified, whereas in a correctional facility or hospital only staff will be notified. Also, if the method is standardized, it is easier to differentiate between emergency and nonemergency signals, which facilitates a smoother reaction when an emergency does occur.

Item 8 is focused on the notification of emergency responders. They are more likely to get the notification of an emergency if a standard protocol exists. This can vary from one occupancy or use to another, but as long as a straightforward, consistent method is used, it will facilitate a quicker response. Note that Section 401.3.2 requires direct contact with the fire department once the fire alarm signal is activated; therefore, no intermediate steps, such as an investigation, are allowed.

As with Item 6, Item 9 requires a specific contact that is familiar with the plan and how the building operates. This information is helpful for the emergency responders in their preplanning activities. Without a specific contact, the process of getting vital information can become much more difficult for the fire department. In a large building or facility, the safety officer or similar person is most appropriate for such a role.

Item 10 requires documentation of the voice/alarm communications system alert tone and preprogrammed voice messages. This provides emergency responders with a better understanding of the information provided to occupants to better assess the appropriate response. Additionally, if conditions in that building change, the plan can be evaluated to see

whether this aspect of the notification system needs to be revised. For instance, if the procedures for evacuation have changed, the voice announcement may need to be revised.

404.2.2 Fire safety plans. Fire safety plans shall include the following:

1. The procedure for reporting a fire or other emergency.
2. The life safety strategy including the following:
 - 2.1. Procedures for notifying occupants, including areas with a private mode alarm system.
 - 2.2. Procedures for occupants under a *defend-in-place* response.
 - 2.3. Procedures for evacuating occupants, including those who need evacuation assistance.
3. Site plans indicating the following:
 - 3.1. The occupancy assembly point.
 - 3.2. The locations of fire hydrants.
 - 3.3. The normal routes of fire department vehicle access.
4. Floor plans identifying the locations of the following:
 - 4.1. Exits.
 - 4.2. Primary evacuation routes.
 - 4.3. Secondary evacuation routes.
 - 4.4. Accessible egress routes.
 - 4.4.1. Areas of refuge.
 - 4.4.2. Exterior areas for assisted rescue.
 - 4.5. Refuge areas associated with *smoke barriers* and *horizontal exits*.
 - 4.6. Manual fire alarm boxes.
 - 4.7. Portable fire extinguishers.
 - 4.8. Occupant-use hose stations.
 - 4.9. Fire alarm annunciators and controls.
5. A list of major fire hazards associated with the normal use and occupancy of the premises, including maintenance and housekeeping procedures.
6. Identification and assignment of personnel responsible for maintenance of systems and equipment installed to prevent or control fires.
7. Identification and assignment of personnel responsible for maintenance, housekeeping and controlling fuel hazard sources.

C This section requires an overall fire safety plan with emphasis on the building and building site layout and hazards. More specifically, information such as the evacuation and relocation aspects of the building layout must be clear, the list of specific hazards associated with normal use of the building must be noted and fire department access road locations must be provided.

This plan also includes identification of specific personnel charged with managing fire protection systems and equipment and with fire prevention duties, such as controlling combustibles on-site. Having specific personnel assigned to perform this duty will work to increase the likelihood of these actions occurring.

The requirements of this plan provide the building owner and occupants a better understanding of how to react in an emergency and how to decrease the likelihood of an emergency occurring. Additionally, this report assists emergency responders during periodic inspections and evaluations of the plans and, more importantly, when responding to an emergency. Generally, buildings that have fairly rigid and well-maintained plans and procedures in place reduce not only the likelihood and magnitude of an incident within the jurisdiction but also the burden to emergency responders.

404.2.3 Lockdown plans. Lockdown plans shall only be permitted where such plans are *approved* by the *fire code official* and are in compliance with Sections 404.2.3.1 and 404.2.3.2.

C This section introduces the requirements for plans and procedures for lockdowns (see commentary to “Lockdown,” Section 202). The section is clear that such plans are allowed only where approved by the fire code official. The code does not require a lockdown plan; however, if a lockdown plan is developed, it must be strictly supervised in order to maintain occupant safety at an acceptable level. Many facilities are adopting procedures that can significantly affect fire and life safety, such as using the fire alarm system to signal a security emergency, locking doors with devices that prevent egress in violation of the provisions of Chapter 10 of the code, and chaining exit discharge doors from the inside to prevent occupants from leaving the building. It is important that plans for security threats do not include procedures that result in violations of life safety and actually increase the hazard to the occupants. The sections that follow are intended to establish the conditions for lockdown plans so that they will not decrease the level of life safety in the event of fire.

404.2.3.1 Lockdown plan contents. Lockdown plans shall include the following:

1. Identification of individuals authorized to issue a lockdown order.
2. Security measures used during normal operations, when the building is occupied, that could adversely affect egress or fire department operations.
3. A description of identified emergency and security threats addressed by the plan, including specific lockdown procedures to be implemented for each threat condition.
4. Means and methods of initiating a lockdown plan for each threat, including:
 - 4.1. The means of notifying occupants of a lockdown event, which shall be distinct from the fire alarm signal.
 - 4.2. Identification of each door or other access point that will be secured.
 - 4.3. A description of the means or methods used to secure doors and other access points.
 - 4.4. A description of how locking means and methods are in compliance with the requirements of this code for egress and accessibility.
5. Procedures for reporting to the fire department any lockdown condition affecting egress or fire department operations.
6. Procedures for determining and reporting the presence or absence of occupants to emergency response agencies during a lockdown.
7. Means for providing two-way communication between a central location and each area subject to being secured during a lockdown.
8. Identification of the prearranged signal for terminating the lockdown.
9. Identification of individuals authorized to issue a lockdown termination order.
10. Procedures for unlocking doors and verifying that the *means of egress* has been returned to normal operations upon termination of the lockdown.
11. Training procedures and frequency of lockdown plan drills.

C The items in this section are intended to more thoroughly address various security means and methods created and implemented in response to various incidents throughout the country. Several such means and methods, while solving security concerns, have unintended consequences regarding emergency egress and fire and life safety systems. Previous editions of this code did not provide the necessary amount of detail for such lockdown plans. It is important to ensure that the plan covers the topics pertinent to the fire code official thoroughly.

A key aspect of a lockdown plan is to identify the organizational structure in charge of making decisions during a lockdown. It is important to understand who is authorized to both initiate and lift a lockdown order. This section requires inclusion of lockdown procedures in the plan. There is often a need for different lockdown plans to address differing threats. The correct procedure during a chemical release incident might differ from the correct procedure during an active shooter incident. The correct procedure for an active shooter in the vicinity of a facility may differ from the procedure for an active shooter on the site of the facility.

A description of the normal status of security, a description of the various threat scenarios addressed by the lockdown plan, and the lockdown procedure for each such scenario are important items to include for review. Such details can impact fire department response and alert fire officials to the life safety implications of any given approach. This is especially true with the methods of locking included in the lockdown plan. There is concern about the proliferation of unlisted locking mechanisms and their potential detrimental impact on the means of egress. This section requires that all means and methods of locking for all access points be described, and that compliance with Chapter 10 requirements be demonstrated. This section also requires procedures as to how lockdown events are communicated with the fire department. In addition, the section requires that a central “command” location be preplanned and that a reliable means of two-way communication be provided in order to keep locked-down occupants apprised of the status of the situation, thus reducing their level of anxiety and the likelihood of panic. Finally, this section requires that the lockdown plan includes training procedures and a requirement as to how frequently drills are necessary.

404.2.3.2 Drills. Lockdown plan drills shall be conducted in accordance with the *approved* plan. Such drills shall not be substituted for fire and evacuation drills required by Section 405.3.

C To utilize lockdown training and the lessons learned from it to the best advantage, training should be conducted on a regular basis to familiarize staff and occupants with the lockdown plan. The training procedures and frequency of drills are established in Section 404.2.3.1. Training should be designed and practiced to reinforce lockdown procedures as adaptive planned responses to stressful and potentially dangerous situations. It is important to note that lockdown training and drills may not be counted toward the required number of fire and evacuation drills so as not to diminish the impact of either (see also commentary, Section 406.3.3).

404.3 Maintenance. Fire safety, evacuation and lockdown plans shall be reviewed or updated annually or as necessitated by changes in staff assignments, occupancy or the physical arrangement of the building.

C In order to be of optimum value to a facility, plans must accurately reflect building conditions. Plans must be reviewed annually or when building changes affecting the instructions or procedures in the fire safety, emergency evacuation or lockdown plan occur. Such a review should prompt an immediate revision and redistribution of the plan to all concerned parties, including emergency response personnel.

404.4 Availability. Fire safety, evacuation and lockdown plans shall be available in the workplace for reference and review by employees, and copies shall be furnished to the *fire code official* for review on request.

C This essentially requires that fire safety, emergency and lockdown plans be easily accessible to building occupants and the fire code official. If the plans are difficult to access, they are less likely to be updated when necessary and are more likely to be lost or forgotten. This places a burden on the emergency responders when planning methods of response and puts the occupants of the building at a higher risk during an emergency. Having the documents readily available makes review and use for training occupants more likely.

404.4.1 Distribution. The fire safety, evacuation and lockdown plans shall be distributed to the tenants and building service employees by the *owner* or *owner's* agent. Tenants shall distribute to their employees applicable parts of the fire safety plan and lockdown plan affecting the employees' actions in the event of a fire or other emergency.

C Fire safety, evacuation and lockdown plans are only effective when all building occupants have been informed of the contents of the plan. In the case of a multiple-tenant building, the plan must address individual tenant spaces. Distribution to all affected occupants is important for a coordinated response to an emergency.

This section requires that the building owner or the owner's agent distribute the plans to all tenants and building service employees. Since the owner or agent of the owner usually does not have direct access to the tenants' employees, the individual tenants would then have the responsibility to distribute the applicable portion of the plan to their employees. This provides for a wider distribution of the responsibility to plan for emergencies and to follow the requirements of the plan.

SECTION 405—EMERGENCY EVACUATION DRILLS

405.1 General. Emergency fire and evacuation drills complying with Sections 405.3 through 405.10 shall be conducted not less than annually where fire safety and evacuation plans are required by Section 403 or where required by the *fire code official*. Lockdown plan drills shall be conducted in accordance with the *approved* plan. Such drills shall not be substituted for fire and evacuation drills required by Section 405.3. Drills shall be designed in cooperation with the local authorities.

C The fire safety and evacuation plans required by Sections 403 and 404 are critical components of life safety strategies in high-density occupancies and extraordinary buildings, such as high-rise buildings. Many times, such plans are developed with the best of intentions and available expertise, but fail to adequately consider certain site- or area-specific conditions. Just as emergency operations and hazardous material response plans require operational drills to verify their continued viability and effectiveness, so too do evacuation plans require periodic implementation to gauge effectiveness in achieving their objectives. Since not all of the occupancies that are required to prepare and maintain fire safety and evacuation plans by Section 403 are required to conduct drills by Table 405.3, this section requires at least an annual drill to exercise the required plan.

An additional benefit of an annual drill is the enhanced recognition by occupants. This is particularly beneficial in buildings such as high-rises that evacuate by selected floors. Given the generally heightened awareness following the events of September 11, 2001, building occupants understandably exhibit a greater reluctance to remain inside a building in the event of an emergency. A complete building evacuation may be unnecessary in a large number of cases and may actually work counter to effective fire-ground operations if uncontrolled. An annual drill in these instances would serve to reinforce the local jurisdiction's preferences regarding evacuation and provide a verification tool that the plan, as developed, is functional, while also providing opportunities for critique and improvement. The use of an annual drill could easily be incorporated with specific additional training to satisfy the requirements in Section 406 as well.

This section and Section 404.2.3.2 clarify that lockdown plan drills must also be conducted but are a separate activity from drills required by this section. Such drills cannot be substituted for the necessary evacuation drills in this section, which address a different hazard related to fire.

405.2 Occupant participation. Emergency fire and evacuation drills shall involve the actual evacuation of occupants to a selected assembly point and shall provide occupants with experience in exiting through required *exits*.

Exceptions:

1. In ambulatory care facilities and Group I-2, the movement of care recipients to a safe area or to the exterior of the building is not required.
2. In Group I-1, Condition 2, the assembly point for residents is permitted to be within an adjacent *smoke compartment*.

3. In Group R-4, actual exiting from emergency escape and rescue openings shall not be required. Opening the emergency escape and rescue openings and signaling for help shall be an acceptable alternative.
4. In Group I-3, Conditions 2 through 5 where a *defend-in-place* response is permitted, the assembly point for detainees is permitted to be within an adjacent *smoke compartment*.
5. In Group I-3, Conditions 2 through 5, movement of detainees is not required to an assembly point where there are security concerns.

C This section provides the verbiage to ensure that fire and evacuation drills involve the actual evacuation of occupants, including actually using the available required exits. The intent is for all occupants to participate, not simply staff.

There are a number of occupancies that have necessary exceptions based on the type and condition of occupants. Specific defend-in-place strategies are in place for ambulatory care facilities; Group I-2; and Group I-1, Condition 2 occupancies. Since R-4 occupancies contain emergency escape openings that could lead to unnecessary injury if used during evacuation drills, occupants can simply open the emergency escape and rescue opening and provide a signal to denote they are properly located. Exceptions 4 and 5 are in recognition of detainee participation in drills for correctional facilities.

405.3 Frequency. Required emergency evacuation drills shall be held at the intervals specified in Table 405.3 or more frequently where necessary to familiarize all occupants with the drill procedure.

C To utilize fire drills and the lessons that they teach to the best of their advantage, drills should be conducted on a regular basis to familiarize both staff and residents with the evacuation plan. The element of surprise is not necessarily of significant benefit in those occupancies where residents may be prone to maladaptive behavior. Drills should be designed and practiced to reinforce relocation or evacuation behaviors as adaptive planned responses to stressful and potentially dangerous situations. Drills should be scheduled so that all staff members on all shifts have an opportunity to participate in them. Practice makes perfect, and when it comes to effective egress, there is no substitute for fire drills at regular intervals so that all occupants are familiar with the plan's details and their particular responsibilities in implementing them. Truly effective drills test the plan by varying conditions and forcing occupants to adapt. Many conditions can combine to affect available safe egress time, and drills should incorporate some allowance for unanticipated conditions, such as delayed detection, rapid fire growth, reduced staffing or poor weather conditions, as may be appropriate for the occupancy. Discovering deficiencies in the evacuation plan should be encouraged, and every opportunity should be taken to improve the plans.

TABLE 405.3—FIRE AND EVACUATION DRILL FREQUENCY AND PARTICIPATION

GROUP OR OCCUPANCY	FREQUENCY	PARTICIPATION
Group A	Quarterly	Staff
Group B ^a	Annually	All occupants
Group B (Ambulatory care facilities)	Quarterly on each shift	Staff
Group B ^a (Clinic, outpatient)	Annually	Staff
Group E	Monthly	All occupants
Group F	Annually	Employees
Group I-1 ^c	Semiannually on each shift	All occupants
Group I-2	Quarterly on each shift	Staff
Group I-3	Quarterly on each shift	Staff
Group I-4	Monthly on each shift	All occupants
Group R-1	Quarterly on each shift	Employees
Group R-2 ^b	Four annually	All occupants
Group R-4 ^c	Semiannually on each shift	All occupants
a. Emergency evacuation drills are required in Group B buildings having an occupant load of 500 or more persons or more than 100 persons above or below the lowest level of exit discharge. b. Emergency evacuation drills in Group R-2 college and university buildings shall be in accordance with Section 403.9.2.1. Other Group R-2 occupancies shall be in accordance with Section 403.9.2.2. c. In Groups I-1 and R-4, see Sections 403.7.1.4 and 403.9.3.4 for additional drills for staff.		

C Table 405.3 provides varying evacuation drill frequencies based on building occupancy. The table also prescribes who should be involved in these drills. The level of participation is based on the type of occupancy. In some cases, this is further qualified by the number or location of such occupants with respect to the lowest level of fire department vehicle access. It

is unreasonable, for example, in a Group A occupancy such as a hospital or correctional facility to require the general public to participate in an evacuation drill. It is potentially dangerous to involve anyone but the employees in such drills.

The overall strategy for these occupancies is to provide a package of relevant information before an emergency and to have the employees facilitate and direct occupants during an emergency. The necessary participants in the drills are related to the overall emergency strategies for those buildings. Note that Group B, E, I-1 assisted living, I-4, R-2 and R-4 occupancies require that everyone be involved. This is related to the fact that occupants are generally able to evacuate. In the case of educational occupancies, drills serve as a learning tool for children to carry through their lives. Schools have generally stressed these drills because of large losses in fires, such as in 1958 at Our Lady of Angels School in Chicago. In terms of Group I-1, I-4 and R-4 facilities, the occupants generally are able to evacuate with some assistance. In Group R-2 occupancies, the occupants are usually able-bodied, and it is in their best interest to be familiar with the egress routes. More occupancy-specific discussion is provided in the commentary for Section 403.

405.4 Leadership. Responsibility for the planning and conduct of drills shall be assigned to competent persons designated to exercise leadership.

C This section requires a focal point in the planning and execution of evacuation drills. Having a single point of contact streamlines the process and provides a necessary leadership role.

405.5 Time. Drills shall be held at unexpected times and under varying conditions to simulate the unusual conditions that occur in case of fire.

Exceptions:

1. In severe climates, the *fire code official* shall have the authority to modify the emergency evacuation drill termination points and frequency.
2. In Groups I-1, I-2, I-3 and R-4, where staff-only emergency evacuation drills are conducted after visiting hours or where care recipients are expected to be asleep, a coded announcement shall be an acceptable alternative to audible alarms.

C If fire and emergency drills are a routine planned occurrence, they will not simulate actual reaction to an emergency but will provide an inaccurate and most likely optimistic outcome; therefore, the drills need to occur at random. Two exceptions are provided. Exception 1 recognizes that in some climates, termination points and frequency of drills may necessarily be modified. For instance, having occupants stand outside in extreme heat or low temperatures may do more harm than any benefit an evacuation drill can provide, especially for care recipients in Group R-4 occupancies. Exception 2 recognizes the higher risk of disturbing care recipients who may be expected to be asleep with loud noises during a staff-only drill; therefore, it allows the use of a coded announcement to alert staff.

405.6 Recordkeeping. Records shall be maintained of required emergency evacuation drills and include the following information:

1. Identity of the person conducting the drill.
2. Date and time of the drill.
3. Notification method used.
4. Employees on duty and participating.
5. Number of occupants evacuated.
6. Special conditions simulated.
7. Problems encountered.
8. Weather conditions when occupants were evacuated.
9. Time required to accomplish complete evacuation.

C Documenting the frequency and efficiency of emergency evacuation drills not only aids the fire code official in verifying that drills complying with these provisions have been performed but may also help administrators identify trends in emergency evacuation drill performance. Accurate records help life safety planners determine the adequacy of their plans. Identifying issues such as problems encountered and weather conditions helps to further determine which elements create the largest delays and why.

405.7 Notification. Where required by the *fire code official*, prior notification of emergency evacuation drills shall be given to the *fire code official*.

C In some cases, the fire code official will want prior notification of evacuation drills because they may need to prepare for such an event. This section provides the fire code official with the authority to require such notification.

405.8 Initiation. Where a *fire alarm system* is provided, emergency evacuation drills shall be initiated by activating the *fire alarm system*.

C To simulate conditions normally experienced during an emergency, the emergency notification procedures, which would include a fire alarm system in many cases, must be used.

405.9 Accountability. As building occupants arrive at the assembly point, efforts shall be made to determine if all occupants have been successfully evacuated or have been accounted for.

Exception: In Group I-2 and ambulatory care facilities, the movement of care recipients to safe areas or to the exterior of the building is not required.

C This requirement is key to the success of evacuation plans. If a method is not available to account for the occupants once evacuation or relocation is complete, search and rescue activities will be more difficult for the emergency responders. Also, it would be difficult to measure the success of the plan. The exception recognizes that Group I-2 occupancies and ambulatory care facilities are not designed to physically evacuate care recipients but instead to defend in place or relocate.

405.10 Recall and reentry. An electrically or mechanically operated signal used to recall occupants after an evacuation shall be separate and distinct from the signal used to initiate the evacuation. The recall signal initiation means shall be manually operated and under the control of the person in charge of the premises or the official in charge of the incident. Persons shall not reenter the premises until authorized to do so by the official in charge.

C This section is primarily aimed at the concern that occupants will be confused if similar signals are used to notify them of an alarm and for reentry. This confusion has the consequences of slowing or even halting evacuation during an actual emergency. Additionally, to make sure that occupants do not go back into the building prematurely, any reentry signal must be operated manually to avoid a situation where it automatically sounds. Finally, this section specifically prohibits reentry until authorization is provided by the official in charge at the scene.

SECTION 406—EMPLOYEE TRAINING AND RESPONSE PROCEDURES

406.1 General. Where fire safety and evacuation plans are required by Section 403, employees shall be trained in fire emergency procedures based on plans prepared in accordance with Section 404.

C In most cases, the success of an evacuation and fire safety plan hinges on the appropriate reactions of the building occupants. The main activity that building occupants must undertake is removing themselves from the hazards. In some cases, fire safety and evacuation plans involve additional actions by the employees of the facility. For instance, in the case of hospitals, the nurses and other hospital staff must relocate care recipients; therefore, specific training is required for those activities.

Additionally, employees must be trained based on the specific fire safety and fire evacuation plans.

406.2 Frequency. Employees shall receive training in the contents of fire safety and evacuation plans and their duties as part of new employee orientation and not less than annually thereafter. Records of training shall be maintained.

C This section requires that employee training occur during new employee indoctrination and annually thereafter. A record of this training must be provided to the fire code official when requested. This section provides a minimum criterion for the training frequency for all occupancies addressed by Chapter 4. Section 403 may require more restrictive training frequencies.

406.3 Employee training program. Employees shall be trained in fire prevention, evacuation and fire safety in accordance with Sections 406.3.1 through 406.3.4.

C This is a general section that requires all employees to be trained in fire prevention, evacuation and fire safety in accordance with the subsections that follow. These provisions are primarily intended as a mechanism to ensure that training occurs, not as a requirement for establishing training criteria.

406.3.1 Fire prevention training. Employees shall be apprised of the fire hazards of the materials and processes to which they are exposed. Each employee shall be instructed in the proper procedures for preventing fires in the conduct of their assigned duties.

C If a fire can be prevented, evacuation and relocation of the occupants will be avoided. Employees must be made aware of the potential hazards related to their particular area of the facility and what can be done to avoid a hazardous situation. Having specific procedures increases the likelihood that proper fire prevention techniques will be followed. Generally, people tend to be unaware of many hazards unless they are alerted to them. An example is the use of space heaters.

406.3.2 Evacuation training. Employees shall be familiarized with the fire alarm and evacuation signals, their assigned duties in the event of an alarm or emergency, evacuation routes, areas of refuge, exterior assembly areas and procedures for evacuation.

C In the event that an emergency does occur, employees must be prepared to assist in the evacuation or relocation of occupants. This training will vary widely from one occupancy type to another. In a high-rise building only some of the occupants will be evacuated at a time, whereas a school will evacuate all occupants at once. This section requires that the training occurs.

406.3.3 Fire safety training. Employees assigned firefighting duties shall be trained to know the locations and proper use of portable fire extinguishers or other manual firefighting equipment and the protective clothing or equipment required for its safe and proper use.

C Any time employees are to take specific action during a fire event, proper training is required. This section holds the building owner or operator responsible for making sure the training occurs.

406.3.4 Emergency lockdown training. Where a facility has a lockdown plan, employees shall be trained on their assigned duties and procedures in the event of an emergency lockdown.

C If a facility has created a lockdown plan, the staff needs to be versed in the plan's contents. Training should be scheduled so that all staff members on all shifts or occupants, as applicable, have an opportunity to participate. Practice makes perfect, and when it comes to effective lockdowns, there is no substitute for training at regular intervals so that all staff and occupants are familiar with the plan's details and their particular responsibilities in implementing it. Drills should be scheduled to test the plan by varying conditions, forcing occupants to adapt and respond instinctively to emergencies without panicking and allowing the staff to identify weaknesses in the preparations.

Many conditions can conspire to affect lockdown success and drills should incorporate some allowance for unanticipated conditions as may be appropriate for the occupancy. Discovering deficiencies in the plan should be encouraged, and every opportunity should be taken to improve the plans (see also commentary, Section 404.2.3.2).

SECTION 407—HAZARD COMMUNICATION

407.1 General. Where operating or closing a hazardous materials storage, use or handling facility subject to permits under Section 5001.5, or where required by the *fire code official*, the provisions of Sections 407.2 through 407.7 shall be applicable.

C This section is specific to buildings and facilities that contain hazardous materials over the permitted amounts listed in Section 105.5. Knowledge related to which hazardous materials are on-site is critical for several reasons. First, it assists emergency responders in knowing what to expect when responding to a scene. Second, it provides emergency responders with an idea of incidents that may occur at a building or facility. Lastly, it provides a better understanding to the occupants of the potential hazards present and how to avoid emergencies.

Facilities that store, use or handle hazardous materials on a large scale generally depend heavily on the actions of the employees to prevent or minimize hazardous materials incidents. Therefore, the occupants play a strong role in the overall protection of the building. These types of requirements would, in the past, have been found only within the hazardous materials section of the code. These provisions have also been located within this chapter because of the nature of the information. These requirements are aimed at preparing both the occupants and the emergency responders. Hazardous materials present a wide range of problems because of the significant variation of properties and reactions; therefore, a reference to these requirements has been included in Chapter 4. These specific provisions are found in Section 5001.

This section states that when subject to permit requirements, hazardous materials must also be addressed. This section has requirements for the submittal of Safety Data Sheets (SDS); the labeling or marking of hazardous materials through placarding and related identification; training; the compilation, where required, of both HMIS and HMMP; and finally, the submittal of a closure plan where a facility is being shut down.

407.2 Safety Data Sheets. Safety Data Sheets (SDS) for all hazardous materials shall be either readily available on the premises as a paper copy, or where *approved*, shall be permitted to be readily retrievable by electronic access.

C SDS provide critical information about individual chemicals and their related hazards. This section requires that these data sheets be readily available on the premises or, where equivalent access can be achieved, as approved by the fire code official, they may be retrievable by electronic means. This allows availability for review by both employees and emergency responders. An appropriate location may be the security room at a facility or perhaps the main office. These sheets can potentially play a role in the response to an emergency. For example, if a chemical is noted on the SDS as being water reactive, depending on the level of water reactivity, applying water to that spill may not be an appropriate response.

407.3 Identification. Individual containers of hazardous materials, cartons or packages shall be marked or labeled in accordance with applicable federal regulations. Buildings, rooms and spaces containing hazardous materials shall be identified by hazard warning signs in accordance with Section 5003.5.

C This section requires two activities related to the identification of hazardous materials. First, chemicals must be specifically labeled. Second, rooms or areas where the materials are located must be specifically labeled. In this case, the code essentially requires placarding as defined in Section 5003.5, which references NFPA 704.

407.4 Training. Persons responsible for the operation of areas in which hazardous materials are stored, dispensed, handled or used shall be familiar with the chemical nature of the materials and the appropriate mitigating actions necessary in the event of a fire, leak or spill. Responsible persons shall be designated and trained to be liaison personnel for the fire department. These persons shall aid the fire department in preplanning emergency responses and identification of where hazardous materials are located, and shall have access to Safety Data Sheets and be knowledgeable in the site emergency response procedures.

C This section requires training specific to the hazards of the materials located and used at a particular building or facility. As noted, the actions taken will vary based on the hazards associated with the materials. Additionally, this section requires specific persons to be designated as points of contact for the fire department. Having specific points of contact is critical because it eases planning and response procedures. These contacts also provide the fire department with specific persons who are more familiar with the hazards of the building or facility. The fire department is charged with responding to many

different businesses within a community. Having specified contacts at facilities containing hazardous materials helps them prepare and respond.

407.5 Hazardous Materials Inventory Statement. Where required by the *fire code official*, each application for a permit shall include a Hazardous Materials Inventory Statement (HMIS) in accordance with Section 5001.5.2.

C An HMIS is further described in Chapter 50 but is essentially a document listing all the hazardous materials found on-site. This documentation includes information such as the type and amount of specific hazards associated with the material and how it is used. All of this information can be very important for emergency planning and preparedness. An HMIS is necessary only if the fire code official specifically requires one. See Appendix H for further information on HMIS and HMMP preparation.

407.6 Hazardous Materials Management Plan. Where required by the *fire code official*, each application for a permit shall include a Hazardous Materials Management Plan (HMMP) in accordance with Section 5001.5.1. The *fire code official* is authorized to accept a similar plan required by other regulations.

C As with the HMIS, an HMMP is necessary only where the fire code official specifically requires one. This document is somewhat different from the HMIS in that it is geared to the layout of the building and the location and use of the hazardous materials. This document provides a better understanding of how the facility operates. This information in turn provides more detailed information to the emergency responders. This plan will also include such information as the location of aisles; the type and location of emergency equipment available; and location of specific shutoff valves and other operating equipment. The detailed requirements for HMMP are located in Chapter 50. See Appendix H for further information on HMIS and HMMP preparation.

407.7 Facility closure plans. The permit holder or applicant shall submit to the *fire code official* a facility closure plan in accordance with Section 5001.6.3 to terminate storage, dispensing, handling or use of hazardous materials.

C It is important for emergency responders to be made aware of the closure of a plant that uses or stores hazardous materials. First, closure means a readjustment in their planning. Second, the extent of the closure must be communicated so that the emergency responders are made aware of hazards that may still be present. Any hazards that are still present may potentially be more dangerous since the facility is now unattended; therefore, the building owner must develop a plan that is acceptable to the fire code official. In some cases, a facility will be only temporarily closed, which would mean maintaining a permit and continuing inspections.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

Egress Design Solutions (Tubbs and Meacham, 2005).

Evacuation Planning Guide for Stadiums. Washington, DC: U.S. Department of Homeland Security, 2008.

Lithium-Ion Battery Fire Statistics: Market Report & Data. (December 20, 2023). <https://gitnux.org/lithium-ion-battery-fire-statistics/>

SFPE Engineering Guide: Fire Safety for Very Tall Buildings (ICC and SFPE 2021).